

4. Backend Discovery & Modernization Analysis

Objective: Comprehensive backend discovery covering architecture, modules, controller/service/repository layering, database schema, API governance, middleware, authentication & authorization, security, performance, dependencies, secrets, and code quality.

Date: 2026-08-07 12:10:49 UTC | **Scope:** shende-shweta/pingcrm (master) — PHP 8.2 / Laravel 11 + Inertia.js + React, Sanctum 4 auth

Executive Summary

EXECUTIVE SUMMARY

The repository is a Laravel 11 PingCRM base whose original CRM surface (Contacts, Users, Organizations, Reports) remains clean and idiomatic, but a large synthetic "IVR Enterprise" legacy subsystem has been grafted on top and it concentrates almost every backend anti-pattern in the catalogue. The IVR layer ships 82 fat single-action controllers, 12 ~373-line "God" services holding mutable `public static` state and hard-coded API keys, and 12 unused repository classes that build SQL by string concatenation. Untyped request data is materialised with `extract()` in 92 files, raw SQL is concatenated directly in controllers and repositories (SQL-injection signature), Eloquent models use `$guarded = []` (mass-assignment wide open), and a `config/ivr_legacy.php` file commits a master API key, a Salesforce client secret and a plaintext password. An 80-route `ivr-legacy` API is exposed with **no authentication middleware, no OpenAPI spec, no versioning and no contract tests**, while a hard-coded `tenant_id = 1` breaks multi-tenant isolation (IDOR). Performance is degraded by ~540 synchronous `sleep(1)` calls on the request path, 420 N+1 accessor methods and zero caching. Database schema and migrations are the one bright spot (indexed FKs, reversible `down()` methods). The overall backend rating is **High Risk**, driven by broken access control, injection/secret exposure and the absence of a real service/data layer.

91

CONTROLLERS /
HANDLERS
SCANNED

92

FILES USING
DYNAMIC-
VARIABLE
PATTERNS

12

SERVICE
CLASSES FOUND
(ALL "GOD")

81

API ENDPOINTS
FOUND

116

INJECTION /
MASS-
ASSIGNMENT
FILES

N/A*

CRITICAL / HIGH
CVES (AUDIT
NOT RUN
OFFLINE)

OVERALL CODEBASE RATING — BACKEND MODERNIZATION

High Risk

Driven by H12 broken access control (80 unauthenticated endpoints + IDOR), H13 injection/secret exposure, H16 committed secrets, and H1–H5 absence of a real service/data layer.

4.1 Benchmark Ratings Summary

One row per hotspot. "Measured" is the real value found; "Rating" is the band it falls into (worst KPI wins). This table is the source for the Overall Codebase Rating banner above.

#	Hotspot	Primary KPI	GOOD	MODERATE	HIGH RISK	Measured	Rating
H1	Dynamic Variable Creation	Dynamic-var-from-input occurrences	0	1–10	>10	<code>extract()</code> in 92 files (12 services × 45, 82 controllers × ≤55)	HIGH RISK
H2	Global Mutable State	Globals / mutable static state	0	1–5	>5	12 <code>public static \$sharedRuntimeCache</code> in God services	HIGH RISK
H3	Direct SQL Outside Data Layer	Data-layer compliance %	>90%	60–90%	<60%	~0% — 83 controllers call <code>DB::</code> directly; repositories unused	HIGH RISK
H4	Static / Singleton Abuse	Business-logic static/singleton classes	0	1–5	>5	12 God services <code>new</code> - instantiated ~4,400× + 5 static helper classes	HIGH RISK
H5	Missing Service Layer	Handlers with inline business logic	<10	10–20	>20	82 IVR controllers with inline rules/SQL	HIGH RISK
H6	API Sprawl	Documented & governed endpoints %	>90%	80–90%	<80%	0% — 80 duplicative <code>ivr-legacy</code> routes, GET+POST on each	HIGH RISK
H7	Missing API Governance	Governance compliance %	100%	90–99%	<90%	0% — no OpenAPI, no versioning, no contract tests	HIGH RISK
H8	Weak Application Architecture	Modules following declared architecture %	>80%	50–80%	<50%	<20% — IVR breaks MVC; logic in controllers/models	HIGH RISK
H9	Missing Module Inventory	Circular dependency count	0	1–3	>3	0 circular deps, but 17 dead files (helpers + repos) unreferenced	MODERATE
H10	Database Schema Weakness	FK indexes % + migrations with rollback %	Both >90%	One <90%	Both <90%	FKs indexed & constrained; all migrations have <code>down()</code>	GOOD
H11	Middleware Weakness	Required middleware present + ordered %	100%	80–99%	<80%	No security headers, no request-ID logging, no explicit CORS	MODERATE
H12	Auth & Authorization Weakness	Protected routes guarded % + hashing algo	100% + bcrypt/argon2	One gap	Both bad	~26% guarded (80 public API routes) + IDOR; hash = bcrypt	HIGH RISK

H13	Backend Security Vulnerabilities	Injection + hardcoded secrets count	0 each	1–3 total	>3 total	SQLi + mass-assignment + <code>APP_DEBUG=true</code> + hardcoded secrets (>3)	HIGH RISK
H14	Performance & Caching Gaps	N+1 patterns found	0	1–5	>5	420 N+1 accessors + 540 <code>sleep(1)</code> + 0 caching	HIGH RISK
H15	Outdated & Vulnerable Dependencies	Critical/High CVEs found	0	1–3	>3	0 observed (audit not run offline); <code>roave/security-advisories</code> present	GOOD
H16	Secrets & Configuration in Source	Hardcoded secrets / .env committed	0	1–2	>2	15+ hardcoded secrets (config/ivr_legacy.php + 12 service keys)	HIGH RISK
H17	Backend Code Quality	Linters in CI + max cyclomatic complexity	Both good	One gap	Both bad	Linters + PHPStan in CI, but level 1 only + massive duplication/dead code	MODERATE
H18	Swallowed Exceptions (additional)	Empty/silent <code>catch</code> blocks (target 0)	0	1–20	>20	~4,400 <code>catch (\Throwable)</code> blocks returning <code>err</code> and continuing	HIGH RISK

H18 (additional) — *Swallowed Exceptions & Silent Failure*. KPI = count of `catch` blocks that suppress the error and return a success-shaped or leaked-message payload; **Good 0 · Moderate 1–20 · High Risk >20**. Chosen because ~4,400 identical `catch (\Throwable $e) { return ["ok"=>false,"err"=>$e->getMessage()]; }` blocks both hide failures from monitoring and leak internal messages to callers — a distinct defect from the injection (H13) and error-config (H13) hotspots.

4.2 Hotspot-by-Hotspot Evidence

H1. Dynamic Variable Creation CRITICAL

Benchmark: dynamic-var-from-input occurrences = `extract()` used in **92 files** → falls in the **High Risk** band (Good 0 · Moderate 1–10 · High Risk >10).

Every legacy controller and God service materialises raw request/payload keys into local variables via `extract()`, so any request field silently creates or shadows a variable.

```
// app/Http/Controllers/Ivr/CallAnalyticsStoreController.php:47-49
$payload = $request->all();
extract($payload);           // any request key becomes a local variable
$service = new CallAnalyticsGodService();
```

```
// app/Legacy/Services/CallAnalyticsGodService.php:15-17
public function orchestrateCallAnalyticsWorkflow1($payload)
```

```
{
    extract($payload);                // unsafe - untyped, untraceable
    self::$sharedRuntimeCache[$tenant_id ?? 1] = $payload;
}
```

Why it matters here: Because `extract()` runs on `$request->all()`, a caller can inject arbitrary variable names (e.g. `$tenantId`, `$apiKey`) and shadow values the method later trusts, and data flow becomes impossible to type-check with PHPStan. The pattern repeats identically across 12 services × 45 methods and 82 controllers, so there is no single place to add validation.

Recommended approach:

1. Introduce Form Request DTOs (extend `App\Http\Requests`) per module action with `rules()` and typed accessors.
2. Delete every `extract($payload)` / `extract($request->all())` and reference validated fields explicitly (`$request->validated()['tenant_id']`).
3. Enable PHPStan `checkExplicitMixed` once dynamic vars are gone; raise `phpstan.neon` level past 1.

92 files affected.

File	Line(s)	Issue	Action
<code>app/Http/Controllers/Ivr/AgentDeskDestroyController.php</code>	48, 61, 74, 87, 100, 113, 126, 139, 152, 165, 178, 191 +43	Dynamic variables materialised from raw request/payload via extract()	Replace with typed Form Request DTO and explicit field access
<code>app/Http/Controllers/Ivr/AgentDeskExportController.php</code>	48, 61, 74, 87, 100, 113, 126, 139, 152, 165, 178, 191 +43	Dynamic variables materialised from raw request/payload via extract()	Replace with typed Form Request DTO and explicit field access
<code>app/Http/Controllers/Ivr/AgentDeskImportController.php</code>	48, 61, 74, 87, 100, 113, 126, 139, 152, 165, 178, 191 +43	Dynamic variables materialised from raw request/payload via extract()	Replace with typed Form Request DTO and explicit field access
<code>app/Http/Controllers/Ivr/AgentDeskIndexController.php</code>	48, 61, 74, 87, 100, 113, 126, 139, 152, 165, 178, 191 +43	Dynamic variables materialised from raw request/payload via extract()	Replace with typed Form Request DTO and explicit field access
<code>app/Http/Controllers/Ivr/AgentDeskStoreController.php</code>	48, 61, 74, 87, 100, 113, 126, 139, 152, 165, 178, 191 +43	Dynamic variables materialised from raw request/payload via extract()	Replace with typed Form Request DTO and explicit field access
<code>app/Http/Controllers/Ivr/AgentDeskSyncController.php</code>	48, 61, 74, 87, 100, 113, 126, 139, 152, 165, 178, 191 +43	Dynamic variables materialised from raw request/payload via extract()	Replace with typed Form Request DTO and explicit field access
<code>app/Http/Controllers/Ivr/AgentDeskUpdateController.php</code>	48, 61, 74, 87, 100, 113, 126, 139, 152, 165, 178, 191 +43	Dynamic variables materialised from raw request/payload via extract()	Replace with typed Form Request DTO and explicit field access
<code>app/Http/Controllers/Ivr/BusinessHoursDestroyController.php</code>	48, 61, 74, 87, 100, 113, 126, 139, 152, 165, 178, 191 +43	Dynamic variables materialised from raw request/payload via extract()	Replace with typed Form Request DTO and explicit field access
<code>app/Http/Controllers/Ivr/BusinessHoursExportController.php</code>	48, 61, 74, 87, 100, 113, 126, 139, 152, 165, 178, 191 +43	Dynamic variables materialised from raw request/payload via extract()	Replace with typed Form Request DTO and explicit field access

app/Http/Controllers/Ivr/BusinessHoursImportController.php	48, 61, 74, 87, 100, 113, 126, 139, 152, 165, 178, 191 +43	Dynamic variables materialised from raw request/payload via extract()	Replace with typed Form Request DTO and explicit field access
app/Http/Controllers/Ivr/BusinessHoursIndexController.php	48, 61, 74, 87, 100, 113, 126, 139, 152, 165, 178, 191 +43	Dynamic variables materialised from raw request/payload via extract()	Replace with typed Form Request DTO and explicit field access
app/Http/Controllers/Ivr/BusinessHoursStoreController.php	48, 61, 74, 87, 100, 113, 126, 139, 152, 165, 178, 191 +43	Dynamic variables materialised from raw request/payload via extract()	Replace with typed Form Request DTO and explicit field access
app/Http/Controllers/Ivr/BusinessHoursSyncController.php	48, 61, 74, 87, 100, 113, 126, 139, 152, 165, 178, 191 +43	Dynamic variables materialised from raw request/payload via extract()	Replace with typed Form Request DTO and explicit field access
app/Http/Controllers/Ivr/BusinessHoursUpdateController.php	48, 61, 74, 87, 100, 113, 126, 139, 152, 165, 178, 191 +43	Dynamic variables materialised from raw request/payload via extract()	Replace with typed Form Request DTO and explicit field access
app/Http/Controllers/Ivr/CallAnalyticsDestroyController.php	48, 61, 74, 87, 100, 113, 126, 139, 152, 165, 178, 191 +43	Dynamic variables materialised from raw request/payload via extract()	Replace with typed Form Request DTO and explicit field access
app/Http/Controllers/Ivr/CallAnalyticsExportController.php	48, 61, 74, 87, 100, 113, 126, 139, 152, 165, 178, 191 +43	Dynamic variables materialised from raw request/payload via extract()	Replace with typed Form Request DTO and explicit field access
app/Http/Controllers/Ivr/CallAnalyticsImportController.php	48, 61, 74, 87, 100, 113, 126, 139, 152, 165, 178, 191 +43	Dynamic variables materialised from raw request/payload via extract()	Replace with typed Form Request DTO and explicit field access
app/Http/Controllers/Ivr/CallAnalyticsIndexController.php	48, 61, 74, 87, 100, 113, 126, 139, 152, 165, 178, 191 +43	Dynamic variables materialised from raw request/payload via extract()	Replace with typed Form Request DTO and explicit field access
app/Http/Controllers/Ivr/CallAnalyticsStoreController.php	48, 61, 74, 87, 100, 113, 126, 139, 152, 165, 178, 191 +43	Dynamic variables materialised from raw request/payload via extract()	Replace with typed Form Request DTO and explicit field access
app/Http/Controllers/Ivr/CallAnalyticsSyncController.php	48, 61, 74, 87, 100, 113, 126, 139, 152, 165, 178, 191 +43	Dynamic variables materialised from raw request/payload via extract()	Replace with typed Form Request DTO and explicit field access
app/Http/Controllers/Ivr/CallAnalyticsUpdateController.php	48, 61, 74, 87, 100, 113, 126, 139, 152, 165, 178, 191 +43	Dynamic variables materialised from raw request/payload via extract()	Replace with typed Form Request DTO and explicit field access
app/Http/Controllers/Ivr/CallFlowDestroyController.php	48, 61, 74, 87, 100, 113, 126, 139, 152, 165, 178, 191 +43	Dynamic variables materialised from raw request/payload via extract()	Replace with typed Form Request DTO and explicit field access
app/Http/Controllers/Ivr/CallFlowExportController.php	48, 61, 74, 87, 100, 113, 126, 139, 152, 165, 178, 191 +43	Dynamic variables materialised from raw request/payload via extract()	Replace with typed Form Request DTO and explicit field access
app/Http/Controllers/Ivr/CallFlowImportController.php	48, 61, 74, 87, 100, 113, 126,	Dynamic variables materialised from raw request/payload via extract()	Replace with typed Form Request DTO

	139, 152, 165, 178, 191 +43		and explicit field access
app/Http/Controllers/Ivr/CallFlowIndexController.php	48, 61, 74, 87, 100, 113, 126, 139, 152, 165, 178, 191 +43	Dynamic variables materialised from raw request/payload via extract()	Replace with typed Form Request DTO and explicit field access
app/Http/Controllers/Ivr/CallFlowStoreController.php	48, 61, 74, 87, 100, 113, 126, 139, 152, 165, 178, 191 +43	Dynamic variables materialised from raw request/payload via extract()	Replace with typed Form Request DTO and explicit field access
app/Http/Controllers/Ivr/CallFlowSyncController.php	48, 61, 74, 87, 100, 113, 126, 139, 152, 165, 178, 191 +43	Dynamic variables materialised from raw request/payload via extract()	Replace with typed Form Request DTO and explicit field access
app/Http/Controllers/Ivr/CallFlowUpdateController.php	48, 61, 74, 87, 100, 113, 126, 139, 152, 165, 178, 191 +43	Dynamic variables materialised from raw request/payload via extract()	Replace with typed Form Request DTO and explicit field access
app/Http/Controllers/Ivr/CallRecordingDestroyController.php	48, 61, 74, 87, 100, 113, 126, 139, 152, 165, 178, 191 +43	Dynamic variables materialised from raw request/payload via extract()	Replace with typed Form Request DTO and explicit field access
app/Http/Controllers/Ivr/CallRecordingExportController.php	48, 61, 74, 87, 100, 113, 126, 139, 152, 165, 178, 191 +43	Dynamic variables materialised from raw request/payload via extract()	Replace with typed Form Request DTO and explicit field access
app/Http/Controllers/Ivr/CallRecordingImportController.php	48, 61, 74, 87, 100, 113, 126, 139, 152, 165, 178, 191 +43	Dynamic variables materialised from raw request/payload via extract()	Replace with typed Form Request DTO and explicit field access
app/Http/Controllers/Ivr/CallRecordingIndexController.php	48, 61, 74, 87, 100, 113, 126, 139, 152, 165, 178, 191 +43	Dynamic variables materialised from raw request/payload via extract()	Replace with typed Form Request DTO and explicit field access
app/Http/Controllers/Ivr/CallRecordingStoreController.php	48, 61, 74, 87, 100, 113, 126, 139, 152, 165, 178, 191 +43	Dynamic variables materialised from raw request/payload via extract()	Replace with typed Form Request DTO and explicit field access
app/Http/Controllers/Ivr/CallRecordingSyncController.php	48, 61, 74, 87, 100, 113, 126, 139, 152, 165, 178, 191 +43	Dynamic variables materialised from raw request/payload via extract()	Replace with typed Form Request DTO and explicit field access
app/Http/Controllers/Ivr/CallRecordingUpdateController.php	48, 61, 74, 87, 100, 113, 126, 139, 152, 165, 178, 191 +43	Dynamic variables materialised from raw request/payload via extract()	Replace with typed Form Request DTO and explicit field access
app/Http/Controllers/Ivr/CallRoutingDestroyController.php	48, 61, 74, 87, 100, 113, 126, 139, 152, 165, 178, 191 +43	Dynamic variables materialised from raw request/payload via extract()	Replace with typed Form Request DTO and explicit field access
app/Http/Controllers/Ivr/CallRoutingExportController.php	48, 61, 74, 87, 100, 113, 126, 139, 152, 165, 178, 191 +43	Dynamic variables materialised from raw request/payload via extract()	Replace with typed Form Request DTO and explicit field access
app/Http/Controllers/Ivr/CallRoutingImportController.php	48, 61, 74, 87, 100, 113, 126, 139, 152, 165, 178, 191 +43	Dynamic variables materialised from raw request/payload via extract()	Replace with typed Form Request DTO and explicit field access

app/Http/Controllers/Ivr/CallRoutingIndexController.php	48, 61, 74, 87, 100, 113, 126, 139, 152, 165, 178, 191 +43	Dynamic variables materialised from raw request/payload via extract()	Replace with typed Form Request DTO and explicit field access
app/Http/Controllers/Ivr/CallRoutingStoreController.php	48, 61, 74, 87, 100, 113, 126, 139, 152, 165, 178, 191 +43	Dynamic variables materialised from raw request/payload via extract()	Replace with typed Form Request DTO and explicit field access
app/Http/Controllers/Ivr/CallRoutingSyncController.php	48, 61, 74, 87, 100, 113, 126, 139, 152, 165, 178, 191 +43	Dynamic variables materialised from raw request/payload via extract()	Replace with typed Form Request DTO and explicit field access
app/Http/Controllers/Ivr/CallRoutingUpdateController.php	48, 61, 74, 87, 100, 113, 126, 139, 152, 165, 178, 191 +43	Dynamic variables materialised from raw request/payload via extract()	Replace with typed Form Request DTO and explicit field access
app/Http/Controllers/Ivr/CustomerProfileIndexController.php	48, 61, 74, 87, 100, 113, 126, 139, 152, 165, 178, 191 +43	Dynamic variables materialised from raw request/payload via extract()	Replace with typed Form Request DTO and explicit field access
app/Http/Controllers/Ivr/CustomerProfileStoreController.php	48, 61, 74, 87, 100, 113, 126, 139, 152, 165, 178, 191 +43	Dynamic variables materialised from raw request/payload via extract()	Replace with typed Form Request DTO and explicit field access
app/Http/Controllers/Ivr/CustomerProfileUpdateController.php	48, 61, 74, 87, 100, 113, 126, 139, 152, 165, 178, 191 +43	Dynamic variables materialised from raw request/payload via extract()	Replace with typed Form Request DTO and explicit field access
app/Http/Controllers/Ivr/DidInventoryDestroyController.php	48, 61, 74, 87, 100, 113, 126, 139, 152, 165, 178, 191 +43	Dynamic variables materialised from raw request/payload via extract()	Replace with typed Form Request DTO and explicit field access
app/Http/Controllers/Ivr/DidInventoryExportController.php	48, 61, 74, 87, 100, 113, 126, 139, 152, 165, 178, 191 +43	Dynamic variables materialised from raw request/payload via extract()	Replace with typed Form Request DTO and explicit field access
app/Http/Controllers/Ivr/DidInventoryImportController.php	48, 61, 74, 87, 100, 113, 126, 139, 152, 165, 178, 191 +43	Dynamic variables materialised from raw request/payload via extract()	Replace with typed Form Request DTO and explicit field access
app/Http/Controllers/Ivr/DidInventoryIndexController.php	48, 61, 74, 87, 100, 113, 126, 139, 152, 165, 178, 191 +43	Dynamic variables materialised from raw request/payload via extract()	Replace with typed Form Request DTO and explicit field access
app/Http/Controllers/Ivr/DidInventoryStoreController.php	48, 61, 74, 87, 100, 113, 126, 139, 152, 165, 178, 191 +43	Dynamic variables materialised from raw request/payload via extract()	Replace with typed Form Request DTO and explicit field access
app/Http/Controllers/Ivr/DidInventorySyncController.php	48, 61, 74, 87, 100, 113, 126, 139, 152, 165, 178, 191 +43	Dynamic variables materialised from raw request/payload via extract()	Replace with typed Form Request DTO and explicit field access
app/Http/Controllers/Ivr/DidInventoryUpdateController.php	48, 61, 74, 87, 100, 113, 126, 139, 152, 165, 178, 191 +43	Dynamic variables materialised from raw request/payload via extract()	Replace with typed Form Request DTO and explicit field access
app/Http/Controllers/Ivr/HistoricalReportsDestroyController.php	48, 61, 74, 87, 100, 113, 126,	Dynamic variables materialised from raw request/payload via extract()	Replace with typed Form Request DTO

	139, 152, 165, 178, 191 +43		and explicit field access
<code>app/Http/Controllers/Ivr/HistoricalReportsExportController.php</code>	48, 61, 74, 87, 100, 113, 126, 139, 152, 165, 178, 191 +43	Dynamic variables materialised from raw request/payload via extract()	Replace with typed Form Request DTO and explicit field access
<code>app/Http/Controllers/Ivr/HistoricalReportsImportController.php</code>	48, 61, 74, 87, 100, 113, 126, 139, 152, 165, 178, 191 +43	Dynamic variables materialised from raw request/payload via extract()	Replace with typed Form Request DTO and explicit field access
<code>app/Http/Controllers/Ivr/HistoricalReportsIndexController.php</code>	48, 61, 74, 87, 100, 113, 126, 139, 152, 165, 178, 191 +43	Dynamic variables materialised from raw request/payload via extract()	Replace with typed Form Request DTO and explicit field access
<code>app/Http/Controllers/Ivr/HistoricalReportsStoreController.php</code>	48, 61, 74, 87, 100, 113, 126, 139, 152, 165, 178, 191 +43	Dynamic variables materialised from raw request/payload via extract()	Replace with typed Form Request DTO and explicit field access
<code>app/Http/Controllers/Ivr/HistoricalReportsSyncController.php</code>	48, 61, 74, 87, 100, 113, 126, 139, 152, 165, 178, 191 +43	Dynamic variables materialised from raw request/payload via extract()	Replace with typed Form Request DTO and explicit field access
<code>app/Http/Controllers/Ivr/HistoricalReportsUpdateController.php</code>	48, 61, 74, 87, 100, 113, 126, 139, 152, 165, 178, 191 +43	Dynamic variables materialised from raw request/payload via extract()	Replace with typed Form Request DTO and explicit field access
<code>app/Http/Controllers/Ivr/LiveMonitoringDestroyController.php</code>	48, 61, 74, 87, 100, 113, 126, 139, 152, 165, 178, 191 +43	Dynamic variables materialised from raw request/payload via extract()	Replace with typed Form Request DTO and explicit field access
<code>app/Http/Controllers/Ivr/LiveMonitoringExportController.php</code>	48, 61, 74, 87, 100, 113, 126, 139, 152, 165, 178, 191 +43	Dynamic variables materialised from raw request/payload via extract()	Replace with typed Form Request DTO and explicit field access
<code>app/Http/Controllers/Ivr/LiveMonitoringImportController.php</code>	48, 61, 74, 87, 100, 113, 126, 139, 152, 165, 178, 191 +43	Dynamic variables materialised from raw request/payload via extract()	Replace with typed Form Request DTO and explicit field access
<code>app/Http/Controllers/Ivr/LiveMonitoringIndexController.php</code>	48, 61, 74, 87, 100, 113, 126, 139, 152, 165, 178, 191 +43	Dynamic variables materialised from raw request/payload via extract()	Replace with typed Form Request DTO and explicit field access
<code>app/Http/Controllers/Ivr/LiveMonitoringStoreController.php</code>	48, 61, 74, 87, 100, 113, 126, 139, 152, 165, 178, 191 +43	Dynamic variables materialised from raw request/payload via extract()	Replace with typed Form Request DTO and explicit field access
<code>app/Http/Controllers/Ivr/LiveMonitoringSyncController.php</code>	48, 61, 74, 87, 100, 113, 126, 139, 152, 165, 178, 191 +43	Dynamic variables materialised from raw request/payload via extract()	Replace with typed Form Request DTO and explicit field access
<code>app/Http/Controllers/Ivr/LiveMonitoringUpdateController.php</code>	48, 61, 74, 87, 100, 113, 126, 139, 152, 165, 178, 191 +43	Dynamic variables materialised from raw request/payload via extract()	Replace with typed Form Request DTO and explicit field access
<code>app/Http/Controllers/Ivr/PromptLibraryDestroyController.php</code>	48, 61, 74, 87, 100, 113, 126, 139, 152, 165, 178, 191 +43	Dynamic variables materialised from raw request/payload via extract()	Replace with typed Form Request DTO and explicit field access

app/Http/Controllers/Ivr/PromptLibraryExportController.php	48, 61, 74, 87, 100, 113, 126, 139, 152, 165, 178, 191 +43	Dynamic variables materialised from raw request/payload via extract()	Replace with typed Form Request DTO and explicit field access
app/Http/Controllers/Ivr/PromptLibraryImportController.php	48, 61, 74, 87, 100, 113, 126, 139, 152, 165, 178, 191 +43	Dynamic variables materialised from raw request/payload via extract()	Replace with typed Form Request DTO and explicit field access
app/Http/Controllers/Ivr/PromptLibraryIndexController.php	48, 61, 74, 87, 100, 113, 126, 139, 152, 165, 178, 191 +43	Dynamic variables materialised from raw request/payload via extract()	Replace with typed Form Request DTO and explicit field access
app/Http/Controllers/Ivr/PromptLibraryStoreController.php	48, 61, 74, 87, 100, 113, 126, 139, 152, 165, 178, 191 +43	Dynamic variables materialised from raw request/payload via extract()	Replace with typed Form Request DTO and explicit field access
app/Http/Controllers/Ivr/PromptLibrarySyncController.php	48, 61, 74, 87, 100, 113, 126, 139, 152, 165, 178, 191 +43	Dynamic variables materialised from raw request/payload via extract()	Replace with typed Form Request DTO and explicit field access
app/Http/Controllers/Ivr/PromptLibraryUpdateController.php	48, 61, 74, 87, 100, 113, 126, 139, 152, 165, 178, 191 +43	Dynamic variables materialised from raw request/payload via extract()	Replace with typed Form Request DTO and explicit field access
app/Http/Controllers/Ivr/QueueManagementDestroyController.php	48, 61, 74, 87, 100, 113, 126, 139, 152, 165, 178, 191 +43	Dynamic variables materialised from raw request/payload via extract()	Replace with typed Form Request DTO and explicit field access
app/Http/Controllers/Ivr/QueueManagementExportController.php	48, 61, 74, 87, 100, 113, 126, 139, 152, 165, 178, 191 +43	Dynamic variables materialised from raw request/payload via extract()	Replace with typed Form Request DTO and explicit field access
app/Http/Controllers/Ivr/QueueManagementImportController.php	48, 61, 74, 87, 100, 113, 126, 139, 152, 165, 178, 191 +43	Dynamic variables materialised from raw request/payload via extract()	Replace with typed Form Request DTO and explicit field access
app/Http/Controllers/Ivr/QueueManagementIndexController.php	48, 61, 74, 87, 100, 113, 126, 139, 152, 165, 178, 191 +43	Dynamic variables materialised from raw request/payload via extract()	Replace with typed Form Request DTO and explicit field access
app/Http/Controllers/Ivr/QueueManagementStoreController.php	48, 61, 74, 87, 100, 113, 126, 139, 152, 165, 178, 191 +43	Dynamic variables materialised from raw request/payload via extract()	Replace with typed Form Request DTO and explicit field access
app/Http/Controllers/Ivr/QueueManagementSyncController.php	48, 61, 74, 87, 100, 113, 126, 139, 152, 165, 178, 191 +43	Dynamic variables materialised from raw request/payload via extract()	Replace with typed Form Request DTO and explicit field access
app/Http/Controllers/Ivr/QueueManagementUpdateController.php	48, 61, 74, 87, 100, 113, 126, 139, 152, 165, 178, 191 +43	Dynamic variables materialised from raw request/payload via extract()	Replace with typed Form Request DTO and explicit field access
app/Legacy/Services/AgentDeskGodService.php	15, 23, 31, 39, 47, 55, 63, 71, 79, 87, 95, 103 +33	Dynamic variables materialised from raw request/payload via extract()	Replace with typed Form Request DTO and explicit field access
app/Legacy/Services/BusinessHoursGodService.php	15, 23, 31, 39, 47, 55, 63, 71, 79, 87, 95, 103 +33	Dynamic variables materialised from raw request/payload via extract()	Replace with typed Form Request DTO

			and explicit field access
<code>app/Legacy/Services/CallAnalyticsGodService.php</code>	15, 23, 31, 39, 47, 55, 63, 71, 79, 87, 95, 103 +33	Dynamic variables materialised from raw request/payload via extract()	Replace with typed Form Request DTO and explicit field access
<code>app/Legacy/Services/CallFlowGodService.php</code>	15, 23, 31, 39, 47, 55, 63, 71, 79, 87, 95, 103 +33	Dynamic variables materialised from raw request/payload via extract()	Replace with typed Form Request DTO and explicit field access
<code>app/Legacy/Services/CallRecordingGodService.php</code>	15, 23, 31, 39, 47, 55, 63, 71, 79, 87, 95, 103 +33	Dynamic variables materialised from raw request/payload via extract()	Replace with typed Form Request DTO and explicit field access
<code>app/Legacy/Services/CallRoutingGodService.php</code>	15, 23, 31, 39, 47, 55, 63, 71, 79, 87, 95, 103 +33	Dynamic variables materialised from raw request/payload via extract()	Replace with typed Form Request DTO and explicit field access
<code>app/Legacy/Services/CustomerProfileGodService.php</code>	15, 23, 31, 39, 47, 55, 63, 71, 79, 87, 95, 103 +33	Dynamic variables materialised from raw request/payload via extract()	Replace with typed Form Request DTO and explicit field access
<code>app/Legacy/Services/DidInventoryGodService.php</code>	15, 23, 31, 39, 47, 55, 63, 71, 79, 87, 95, 103 +33	Dynamic variables materialised from raw request/payload via extract()	Replace with typed Form Request DTO and explicit field access
<code>app/Legacy/Services/HistoricalReportsGodService.php</code>	15, 23, 31, 39, 47, 55, 63, 71, 79, 87, 95, 103 +33	Dynamic variables materialised from raw request/payload via extract()	Replace with typed Form Request DTO and explicit field access
<code>app/Legacy/Services/LiveMonitoringGodService.php</code>	15, 23, 31, 39, 47, 55, 63, 71, 79, 87, 95, 103 +33	Dynamic variables materialised from raw request/payload via extract()	Replace with typed Form Request DTO and explicit field access
<code>app/Legacy/Services/PromptLibraryGodService.php</code>	15, 23, 31, 39, 47, 55, 63, 71, 79, 87, 95, 103 +33	Dynamic variables materialised from raw request/payload via extract()	Replace with typed Form Request DTO and explicit field access
<code>app/Legacy/Services/QueueManagementGodService.php</code>	15, 23, 31, 39, 47, 55, 63, 71, 79, 87, 95, 103 +33	Dynamic variables materialised from raw request/payload via extract()	Replace with typed Form Request DTO and explicit field access

H2. Global Mutable State HIGH

Benchmark: mutable static state holding business data = **12 God services** with `public static $sharedRuntimeCache` → falls in the **High Risk** band (Good 0 · Moderate 1–5 · High Risk >5).

```
// app/Legacy/Services/CallAnalyticsGodService.php:10-11
public static $sharedRuntimeCache = []; // mutable global-ish state
private $apiKey = "LEGACY_IVR_KEY_2082"; // hard-coded secret
```

```
// app/Legacy/Services/CallAnalyticsGodService.php:16-18
extract($payload);
```

```
self::$sharedRuntimeCache[$tenant_id ?? 1] = $payload; // cross-request leakage
```

Why it matters here: `public static` arrays persist for the life of the worker process (Octane/queue workers especially), so payloads from one tenant's request remain readable by the next request handled by the same worker — a cross-request data-leakage and correctness hazard. It also makes the services impossible to unit-test in isolation because state carries between test cases.

Recommended approach:

1. Remove `public static $sharedRuntimeCache` ; if caching is needed, inject Laravel's `CacheRepository` with tenant-scoped keys.
2. Register each service in the container and resolve per-request (`scoped()` binding) instead of `new` .
3. Add a regression test asserting no static state survives between two service invocations.

12 files affected.

File	Line(s)	Issue	Action
<code>app/Legacy/Services/AgentDeskGodService.php</code>	10	Mutable static state holds per-request business data (cross-request leakage)	Replace with injected scoped cache/service; remove static property
<code>app/Legacy/Services/BusinessHoursGodService.php</code>	10	Mutable static state holds per-request business data (cross-request leakage)	Replace with injected scoped cache/service; remove static property
<code>app/Legacy/Services/CallAnalyticsGodService.php</code>	10	Mutable static state holds per-request business data (cross-request leakage)	Replace with injected scoped cache/service; remove static property
<code>app/Legacy/Services/CallFlowGodService.php</code>	10	Mutable static state holds per-request business data (cross-request leakage)	Replace with injected scoped cache/service; remove static property
<code>app/Legacy/Services/CallRecordingGodService.php</code>	10	Mutable static state holds per-request business data (cross-request leakage)	Replace with injected scoped cache/service; remove static property
<code>app/Legacy/Services/CallRoutingGodService.php</code>	10	Mutable static state holds per-request business data (cross-request leakage)	Replace with injected scoped cache/service; remove static property
<code>app/Legacy/Services/CustomerProfileGodService.php</code>	10	Mutable static state holds per-request business data (cross-request leakage)	Replace with injected scoped cache/service; remove static property
<code>app/Legacy/Services/DidInventoryGodService.php</code>	10	Mutable static state holds per-request business data (cross-request leakage)	Replace with injected scoped cache/service; remove static property
<code>app/Legacy/Services/HistoricalReportsGodService.php</code>	10	Mutable static state holds per-request business data (cross-request leakage)	Replace with injected scoped cache/service; remove static property
<code>app/Legacy/Services/LiveMonitoringGodService.php</code>	10	Mutable static state holds per-request business data (cross-request leakage)	Replace with injected scoped cache/service; remove static property
<code>app/Legacy/Services/PromptLibraryGodService.php</code>	10	Mutable static state holds per-request business data (cross-request leakage)	Replace with injected scoped cache/service; remove static property
<code>app/Legacy/Services/QueueManagementGodService.php</code>	10	Mutable static state holds per-request business data (cross-request leakage)	Replace with injected scoped cache/service; remove static property

H3. Direct SQL / ORM Outside a Data Layer CRITICAL

Benchmark: data-layer compliance = ~0% — 83 controllers issue `DB::` calls directly while the 12

`Repositories/Legacy` classes are never referenced → **High Risk** band (Good >90% · Moderate 60–90% · High Risk <60%).

```
// app/Http/Controllers/Ivr/CallAnalyticsStoreController.php:28
$rows = DB::select("select * from ivr_call_analyticss where name like '%".$q."%' and tenant_id = ".$this->tenantId);
```

```
// app/Repositories/Legacy/CallAnalyticsRepository.php:11-16 (exists but
unused)
$sql = "SELECT * FROM ivr_call_analyticss WHERE tenant_id = " . (int)
$tenantId;
if ($filter) {
    $sql .= " AND name LIKE '%" . $filter . "%'"; // SQLi pattern
}
return DB::select($sql);
```

Why it matters here: Persistence logic is scattered across 83 controller files, so changing a table or adding tenant scoping means editing dozens of handlers; and because the queries are built by concatenation they are simultaneously the SQL-injection surface (see H13). A repository layer already exists (`app/Repositories/Legacy`) but is dead code — the intended data layer was never wired in.

Recommended approach:

1. Move all `DB::` access behind the existing `Repositories/Legacy/*Repository` classes, rewritten with parameter binding / query builder.
2. Inject repositories into services (not controllers); controllers call services only.
3. Add an architecture test (e.g. Pest arch or a PHPStan rule) forbidding `DB::` and `\Illuminate\Support\Facades\DB` inside `App\Http\Controllers`.

83 files affected.

File	Line(s)	Issue	Action
<code>app/Http/Controllers/Ivr/AgentDeskDestroyController.php</code>	28	Raw DB access issued directly from controller (no data layer)	Move query into a Repository with bound parameters; call via service
<code>app/Http/Controllers/Ivr/AgentDeskExportController.php</code>	28	Raw DB access issued directly from controller (no data layer)	Move query into a Repository with bound parameters; call via service
<code>app/Http/Controllers/Ivr/AgentDeskImportController.php</code>	28	Raw DB access issued directly from controller (no data layer)	Move query into a Repository with bound parameters; call via service
<code>app/Http/Controllers/Ivr/AgentDeskIndexController.php</code>	28	Raw DB access issued directly from controller (no data layer)	Move query into a Repository with bound parameters; call via service
<code>app/Http/Controllers/Ivr/AgentDeskStoreController.php</code>	28	Raw DB access issued directly from controller (no data layer)	Move query into a Repository with bound parameters; call via service
<code>app/Http/Controllers/Ivr/AgentDeskSyncController.php</code>	28	Raw DB access issued directly from controller (no data layer)	Move query into a Repository with bound parameters; call via service
<code>app/Http/Controllers/Ivr/AgentDeskUpdateController.php</code>	28	Raw DB access issued directly from controller (no data layer)	Move query into a Repository with bound parameters; call via service
<code>app/Http/Controllers/Ivr/BusinessHoursDestroyController.php</code>	28	Raw DB access issued directly from controller (no data layer)	Move query into a Repository with bound parameters; call via service
<code>app/Http/Controllers/Ivr/BusinessHoursExportController.php</code>	28	Raw DB access issued directly from controller (no data layer)	Move query into a Repository with bound parameters; call via service

[illegible]

app/Http/Controllers/Ivr/CallRecordingDestroyController.php	28	Raw DB access issued directly from controller (no data layer)	Move query into a Repository with bound parameters; call via service
app/Http/Controllers/Ivr/CallRecordingExportController.php	28	Raw DB access issued directly from controller (no data layer)	Move query into a Repository with bound parameters; call via service
app/Http/Controllers/Ivr/CallRecordingImportController.php	28	Raw DB access issued directly from controller (no data layer)	Move query into a Repository with bound parameters; call via service
app/Http/Controllers/Ivr/CallRecordingIndexController.php	28	Raw DB access issued directly from controller (no data layer)	Move query into a Repository with bound parameters; call via service
app/Http/Controllers/Ivr/CallRecordingStoreController.php	28	Raw DB access issued directly from controller (no data layer)	Move query into a Repository with bound parameters; call via service
app/Http/Controllers/Ivr/CallRecordingSyncController.php	28	Raw DB access issued directly from controller (no data layer)	Move query into a Repository with bound parameters; call via service
app/Http/Controllers/Ivr/CallRecordingUpdateController.php	28	Raw DB access issued directly from controller (no data layer)	Move query into a Repository with bound parameters; call via service
app/Http/Controllers/Ivr/CallRoutingDestroyController.php	28	Raw DB access issued directly from controller (no data layer)	Move query into a Repository with bound parameters; call via service
app/Http/Controllers/Ivr/CallRoutingExportController.php	28	Raw DB access issued directly from controller (no data layer)	Move query into a Repository with bound parameters; call via service
app/Http/Controllers/Ivr/CallRoutingImportController.php	28	Raw DB access issued directly from controller (no data layer)	Move query into a Repository with bound parameters; call via service
app/Http/Controllers/Ivr/CallRoutingIndexController.php	28	Raw DB access issued directly from controller (no data layer)	Move query into a Repository with bound parameters; call via service
app/Http/Controllers/Ivr/CallRoutingStoreController.php	28	Raw DB access issued directly from controller (no data layer)	Move query into a Repository with bound parameters; call via service
app/Http/Controllers/Ivr/CallRoutingSyncController.php	28	Raw DB access issued directly from controller (no data layer)	Move query into a Repository with bound parameters; call via service
app/Http/Controllers/Ivr/CallRoutingUpdateController.php	28	Raw DB access issued directly from controller (no data layer)	Move query into a Repository with bound parameters; call via service
app/Http/Controllers/Ivr/Concerns/LoadsIvrModuleData.php	82, 110, 140, 168, 185, 204	Raw DB access issued directly from controller (no data layer)	Move query into a Repository with bound parameters; call via service
app/Http/Controllers/Ivr/CustomerProfileIndexController.php	28	Raw DB access issued directly from controller (no data layer)	Move query into a Repository with bound parameters; call via service
app/Http/Controllers/Ivr/CustomerProfileStoreController.php	28	Raw DB access issued directly from controller (no data layer)	Move query into a Repository with bound parameters; call via service
app/Http/Controllers/Ivr/CustomerProfileUpdateController.php	28	Raw DB access issued directly from controller (no data layer)	Move query into a Repository with bound parameters; call via service
app/Http/Controllers/Ivr/DidInventoryDestroyController.php	28	Raw DB access issued directly from controller (no data layer)	Move query into a Repository with bound parameters; call via service

app/Http/Controllers/Ivr/DidInventoryExportController.php	28	Raw DB access issued directly from controller (no data layer)	Move query into a Repository with bound parameters; call via service
app/Http/Controllers/Ivr/DidInventoryImportController.php	28	Raw DB access issued directly from controller (no data layer)	Move query into a Repository with bound parameters; call via service
app/Http/Controllers/Ivr/DidInventoryIndexController.php	28	Raw DB access issued directly from controller (no data layer)	Move query into a Repository with bound parameters; call via service
app/Http/Controllers/Ivr/DidInventoryStoreController.php	28	Raw DB access issued directly from controller (no data layer)	Move query into a Repository with bound parameters; call via service
app/Http/Controllers/Ivr/DidInventorySyncController.php	28	Raw DB access issued directly from controller (no data layer)	Move query into a Repository with bound parameters; call via service
app/Http/Controllers/Ivr/DidInventoryUpdateController.php	28	Raw DB access issued directly from controller (no data layer)	Move query into a Repository with bound parameters; call via service
app/Http/Controllers/Ivr/HistoricalReportsDestroyController.php	28	Raw DB access issued directly from controller (no data layer)	Move query into a Repository with bound parameters; call via service
app/Http/Controllers/Ivr/HistoricalReportsExportController.php	28	Raw DB access issued directly from controller (no data layer)	Move query into a Repository with bound parameters; call via service
app/Http/Controllers/Ivr/HistoricalReportsImportController.php	28	Raw DB access issued directly from controller (no data layer)	Move query into a Repository with bound parameters; call via service
app/Http/Controllers/Ivr/HistoricalReportsIndexController.php	28	Raw DB access issued directly from controller (no data layer)	Move query into a Repository with bound parameters; call via service
app/Http/Controllers/Ivr/HistoricalReportsStoreController.php	28	Raw DB access issued directly from controller (no data layer)	Move query into a Repository with bound parameters; call via service
app/Http/Controllers/Ivr/HistoricalReportsSyncController.php	28	Raw DB access issued directly from controller (no data layer)	Move query into a Repository with bound parameters; call via service
app/Http/Controllers/Ivr/HistoricalReportsUpdateController.php	28	Raw DB access issued directly from controller (no data layer)	Move query into a Repository with bound parameters; call via service
app/Http/Controllers/Ivr/IvrHubController.php	68, 77, 92, 132, 148, 167, 204, 221, 227, 232, 240, 258 +4	Raw DB access issued directly from controller (no data layer)	Move query into a Repository with bound parameters; call via service
app/Http/Controllers/Ivr/LiveMonitoringDestroyController.php	28	Raw DB access issued directly from controller (no data layer)	Move query into a Repository with bound parameters; call via service
app/Http/Controllers/Ivr/LiveMonitoringExportController.php	28	Raw DB access issued directly from controller (no data layer)	Move query into a Repository with bound parameters; call via service
app/Http/Controllers/Ivr/LiveMonitoringImportController.php	28	Raw DB access issued directly from controller (no data layer)	Move query into a Repository with bound parameters; call via service
app/Http/Controllers/Ivr/LiveMonitoringIndexController.php	28	Raw DB access issued directly from controller (no data layer)	Move query into a Repository with bound parameters; call via service
app/Http/Controllers/Ivr/LiveMonitoringStoreController.php	28	Raw DB access issued directly from controller	Move query into a Repository with bound

		(no data layer)	parameters; call via service
<code>app/Http/Controllers/Ivr/LiveMonitoringSyncController.php</code>	28	Raw DB access issued directly from controller (no data layer)	Move query into a Repository with bound parameters; call via service
<code>app/Http/Controllers/Ivr/LiveMonitoringUpdateController.php</code>	28	Raw DB access issued directly from controller (no data layer)	Move query into a Repository with bound parameters; call via service
<code>app/Http/Controllers/Ivr/PromptLibraryDestroyController.php</code>	28	Raw DB access issued directly from controller (no data layer)	Move query into a Repository with bound parameters; call via service
<code>app/Http/Controllers/Ivr/PromptLibraryExportController.php</code>	28	Raw DB access issued directly from controller (no data layer)	Move query into a Repository with bound parameters; call via service
<code>app/Http/Controllers/Ivr/PromptLibraryImportController.php</code>	28	Raw DB access issued directly from controller (no data layer)	Move query into a Repository with bound parameters; call via service
<code>app/Http/Controllers/Ivr/PromptLibraryIndexController.php</code>	28	Raw DB access issued directly from controller (no data layer)	Move query into a Repository with bound parameters; call via service
<code>app/Http/Controllers/Ivr/PromptLibraryStoreController.php</code>	28	Raw DB access issued directly from controller (no data layer)	Move query into a Repository with bound parameters; call via service
<code>app/Http/Controllers/Ivr/PromptLibrarySyncController.php</code>	28	Raw DB access issued directly from controller (no data layer)	Move query into a Repository with bound parameters; call via service
<code>app/Http/Controllers/Ivr/PromptLibraryUpdateController.php</code>	28	Raw DB access issued directly from controller (no data layer)	Move query into a Repository with bound parameters; call via service
<code>app/Http/Controllers/Ivr/QueueManagementDestroyController.php</code>	28	Raw DB access issued directly from controller (no data layer)	Move query into a Repository with bound parameters; call via service
<code>app/Http/Controllers/Ivr/QueueManagementExportController.php</code>	28	Raw DB access issued directly from controller (no data layer)	Move query into a Repository with bound parameters; call via service
<code>app/Http/Controllers/Ivr/QueueManagementImportController.php</code>	28	Raw DB access issued directly from controller (no data layer)	Move query into a Repository with bound parameters; call via service
<code>app/Http/Controllers/Ivr/QueueManagementIndexController.php</code>	28	Raw DB access issued directly from controller (no data layer)	Move query into a Repository with bound parameters; call via service
<code>app/Http/Controllers/Ivr/QueueManagementStoreController.php</code>	28	Raw DB access issued directly from controller (no data layer)	Move query into a Repository with bound parameters; call via service
<code>app/Http/Controllers/Ivr/QueueManagementSyncController.php</code>	28	Raw DB access issued directly from controller (no data layer)	Move query into a Repository with bound parameters; call via service
<code>app/Http/Controllers/Ivr/QueueManagementUpdateController.php</code>	28	Raw DB access issued directly from controller (no data layer)	Move query into a Repository with bound parameters; call via service
<code>app/Http/Controllers/ReportsController.php</code>	61, 77, 98, 115, 166	Raw DB access issued directly from controller (no data layer)	Move query into a Repository with bound parameters; call via service

H4. Static Methods & Singleton Abuse HIGH

Benchmark: business-logic static/singleton classes = **12 God services instantiated ~4,400×** plus **5 static-only helper classes** → **High Risk** band (Good 0 · Moderate 1–5 · High Risk >5).

```
// app/Http/Controllers/Ivr/CallAnalyticsStoreController.php:48-49  (×55
methods × 82 controllers)
$service = new CallAnalyticsGodService();
$service->orchestrateCallAnalyticsWorkflow1($payload);
```

```
// app/Legacy/Helpers/LegacyIvrCrypto.php:7-12  (all-static, 5 files × ~113
methods)
public static function transform1($value)
{
    if ($value === null) { return ""; }
    return (string) $value . "_2130_1";
}
```

Why it matters here: Services are created with **new** at ~4,400 call sites rather than resolved from the container, so they cannot be mocked, decorated, or swapped, and constructor dependencies (config, cache, logger) cannot be injected. The **LegacyIvr*** helper classes are pure static and — as H9 shows — entirely unreferenced dead weight.

Recommended approach:

1. Bind the 12 services in **AppServiceProvider** and type-hint them in controller **__invoke** signatures for auto-injection.
2. Collapse the 5 static helper classes into a small set of injectable, tested utility services (or delete the unused ones — see H9).
3. Forbid **new *GodService()** via a lint/arch rule after DI migration.

80 files affected.

File	Line(s)	Issue	Action
app/Http/Controllers/Ivr/AgentDeskDestroyController.php	25, 49, 62, 75, 88, 101, 114, 127, 140, 153, 166, 179 +44	Business service instantiated with new instead of dependency injection	Bind service in container and inject via constructor/__invoke
app/Http/Controllers/Ivr/AgentDeskExportController.php	25, 49, 62, 75, 88, 101, 114, 127, 140, 153, 166, 179 +44	Business service instantiated with new instead of dependency injection	Bind service in container and inject via constructor/__invoke
app/Http/Controllers/Ivr/AgentDeskImportController.php	25, 49, 62, 75, 88, 101, 114, 127, 140, 153, 166, 179 +44	Business service instantiated with new instead of dependency injection	Bind service in container and inject via constructor/__invoke
app/Http/Controllers/Ivr/AgentDeskIndexController.php	25, 49, 62, 75, 88, 101, 114, 127, 140, 153, 166, 179 +44	Business service instantiated with new instead of dependency injection	Bind service in container and inject via constructor/__invoke
app/Http/Controllers/Ivr/AgentDeskStoreController.php	25, 49, 62, 75, 88, 101, 114, 127, 140, 153, 166, 179 +44	Business service instantiated with new instead of dependency injection	Bind service in container and inject via constructor/__invoke

[illegible]

	127, 140, 153, 166, 179 +44	instead of dependency injection	
app/Http/Controllers/Ivr/CallAnalyticsUpdateController.php	25, 49, 62, 75, 88, 101, 114, 127, 140, 153, 166, 179 +44	Business service instantiated with new instead of dependency injection	Bind service in container and inject via constructor/ __invoke
app/Http/Controllers/Ivr/CallFlowDestroyController.php	25, 49, 62, 75, 88, 101, 114, 127, 140, 153, 166, 179 +44	Business service instantiated with new instead of dependency injection	Bind service in container and inject via constructor/ __invoke
app/Http/Controllers/Ivr/CallFlowExportController.php	25, 49, 62, 75, 88, 101, 114, 127, 140, 153, 166, 179 +44	Business service instantiated with new instead of dependency injection	Bind service in container and inject via constructor/ __invoke
app/Http/Controllers/Ivr/CallFlowImportController.php	25, 49, 62, 75, 88, 101, 114, 127, 140, 153, 166, 179 +44	Business service instantiated with new instead of dependency injection	Bind service in container and inject via constructor/ __invoke
app/Http/Controllers/Ivr/CallFlowIndexController.php	25, 49, 62, 75, 88, 101, 114, 127, 140, 153, 166, 179 +44	Business service instantiated with new instead of dependency injection	Bind service in container and inject via constructor/ __invoke
app/Http/Controllers/Ivr/CallFlowStoreController.php	25, 49, 62, 75, 88, 101, 114, 127, 140, 153, 166, 179 +44	Business service instantiated with new instead of dependency injection	Bind service in container and inject via constructor/ __invoke
app/Http/Controllers/Ivr/CallFlowSyncController.php	25, 49, 62, 75, 88, 101, 114, 127, 140, 153, 166, 179 +44	Business service instantiated with new instead of dependency injection	Bind service in container and inject via constructor/ __invoke
app/Http/Controllers/Ivr/CallFlowUpdateController.php	25, 49, 62, 75, 88, 101, 114, 127, 140, 153, 166, 179 +44	Business service instantiated with new instead of dependency injection	Bind service in container and inject via constructor/ __invoke
app/Http/Controllers/Ivr/CallRecordingDestroyController.php	25, 49, 62, 75, 88, 101, 114, 127, 140, 153, 166, 179 +44	Business service instantiated with new instead of dependency injection	Bind service in container and inject via constructor/ __invoke
app/Http/Controllers/Ivr/CallRecordingExportController.php	25, 49, 62, 75, 88, 101, 114, 127, 140, 153, 166, 179 +44	Business service instantiated with new instead of dependency injection	Bind service in container and inject via constructor/ __invoke
app/Http/Controllers/Ivr/CallRecordingImportController.php	25, 49, 62, 75, 88, 101, 114, 127, 140, 153, 166, 179 +44	Business service instantiated with new instead of dependency injection	Bind service in container and inject via constructor/ __invoke
app/Http/Controllers/Ivr/CallRecordingIndexController.php	25, 49, 62, 75, 88, 101, 114, 127, 140, 153, 166, 179 +44	Business service instantiated with new instead of dependency injection	Bind service in container and inject via constructor/ __invoke
app/Http/Controllers/Ivr/CallRecordingStoreController.php	25, 49, 62, 75, 88, 101, 114, 127, 140, 153, 166, 179 +44	Business service instantiated with new instead of dependency injection	Bind service in container and inject via constructor/ __invoke
app/Http/Controllers/Ivr/CallRecordingSyncController.php	25, 49, 62, 75, 88, 101, 114, 127, 140, 153, 166, 179 +44	Business service instantiated with new instead of dependency injection	Bind service in container and inject via constructor/ __invoke

app/Http/Controllers/Ivr/CallRecordingUpdateController.php	25, 49, 62, 75, 88, 101, 114, 127, 140, 153, 166, 179 +44	Business service instantiated with new instead of dependency injection	Bind service in container and inject via constructor/ __invoke
app/Http/Controllers/Ivr/CallRoutingDestroyController.php	25, 49, 62, 75, 88, 101, 114, 127, 140, 153, 166, 179 +44	Business service instantiated with new instead of dependency injection	Bind service in container and inject via constructor/ __invoke
app/Http/Controllers/Ivr/CallRoutingExportController.php	25, 49, 62, 75, 88, 101, 114, 127, 140, 153, 166, 179 +44	Business service instantiated with new instead of dependency injection	Bind service in container and inject via constructor/ __invoke
app/Http/Controllers/Ivr/CallRoutingImportController.php	25, 49, 62, 75, 88, 101, 114, 127, 140, 153, 166, 179 +44	Business service instantiated with new instead of dependency injection	Bind service in container and inject via constructor/ __invoke
app/Http/Controllers/Ivr/CallRoutingIndexController.php	25, 49, 62, 75, 88, 101, 114, 127, 140, 153, 166, 179 +44	Business service instantiated with new instead of dependency injection	Bind service in container and inject via constructor/ __invoke
app/Http/Controllers/Ivr/CallRoutingStoreController.php	25, 49, 62, 75, 88, 101, 114, 127, 140, 153, 166, 179 +44	Business service instantiated with new instead of dependency injection	Bind service in container and inject via constructor/ __invoke
app/Http/Controllers/Ivr/CallRoutingSyncController.php	25, 49, 62, 75, 88, 101, 114, 127, 140, 153, 166, 179 +44	Business service instantiated with new instead of dependency injection	Bind service in container and inject via constructor/ __invoke
app/Http/Controllers/Ivr/CallRoutingUpdateController.php	25, 49, 62, 75, 88, 101, 114, 127, 140, 153, 166, 179 +44	Business service instantiated with new instead of dependency injection	Bind service in container and inject via constructor/ __invoke
app/Http/Controllers/Ivr/CustomerProfileIndexController.php	25, 49, 62, 75, 88, 101, 114, 127, 140, 153, 166, 179 +44	Business service instantiated with new instead of dependency injection	Bind service in container and inject via constructor/ __invoke
app/Http/Controllers/Ivr/CustomerProfileStoreController.php	25, 49, 62, 75, 88, 101, 114, 127, 140, 153, 166, 179 +44	Business service instantiated with new instead of dependency injection	Bind service in container and inject via constructor/ __invoke
app/Http/Controllers/Ivr/CustomerProfileUpdateController.php	25, 49, 62, 75, 88, 101, 114, 127, 140, 153, 166, 179 +44	Business service instantiated with new instead of dependency injection	Bind service in container and inject via constructor/ __invoke
app/Http/Controllers/Ivr/DidInventoryDestroyController.php	25, 49, 62, 75, 88, 101, 114, 127, 140, 153, 166, 179 +44	Business service instantiated with new instead of dependency injection	Bind service in container and inject via constructor/ __invoke
app/Http/Controllers/Ivr/DidInventoryExportController.php	25, 49, 62, 75, 88, 101, 114, 127, 140, 153, 166, 179 +44	Business service instantiated with new instead of dependency injection	Bind service in container and inject via constructor/ __invoke
app/Http/Controllers/Ivr/DidInventoryImportController.php	25, 49, 62, 75, 88, 101, 114, 127, 140, 153, 166, 179 +44	Business service instantiated with new instead of dependency injection	Bind service in container and inject via constructor/ __invoke
app/Http/Controllers/Ivr/DidInventoryIndexController.php	25, 49, 62, 75, 88, 101, 114,	Business service instantiated with new	Bind service in container and inject via constructor/ __invoke

	127, 140, 153, 166, 179 +44	instead of dependency injection	
app/Http/Controllers/Ivr/DidInventoryStoreController.php	25, 49, 62, 75, 88, 101, 114, 127, 140, 153, 166, 179 +44	Business service instantiated with new instead of dependency injection	Bind service in container and inject via constructor/ __invoke
app/Http/Controllers/Ivr/DidInventorySyncController.php	25, 49, 62, 75, 88, 101, 114, 127, 140, 153, 166, 179 +44	Business service instantiated with new instead of dependency injection	Bind service in container and inject via constructor/ __invoke
app/Http/Controllers/Ivr/DidInventoryUpdateController.php	25, 49, 62, 75, 88, 101, 114, 127, 140, 153, 166, 179 +44	Business service instantiated with new instead of dependency injection	Bind service in container and inject via constructor/ __invoke
app/Http/Controllers/Ivr/HistoricalReportsDestroyController.php	25, 49, 62, 75, 88, 101, 114, 127, 140, 153, 166, 179 +44	Business service instantiated with new instead of dependency injection	Bind service in container and inject via constructor/ __invoke
app/Http/Controllers/Ivr/HistoricalReportsExportController.php	25, 49, 62, 75, 88, 101, 114, 127, 140, 153, 166, 179 +44	Business service instantiated with new instead of dependency injection	Bind service in container and inject via constructor/ __invoke
app/Http/Controllers/Ivr/HistoricalReportsImportController.php	25, 49, 62, 75, 88, 101, 114, 127, 140, 153, 166, 179 +44	Business service instantiated with new instead of dependency injection	Bind service in container and inject via constructor/ __invoke
app/Http/Controllers/Ivr/HistoricalReportsIndexController.php	25, 49, 62, 75, 88, 101, 114, 127, 140, 153, 166, 179 +44	Business service instantiated with new instead of dependency injection	Bind service in container and inject via constructor/ __invoke
app/Http/Controllers/Ivr/HistoricalReportsStoreController.php	25, 49, 62, 75, 88, 101, 114, 127, 140, 153, 166, 179 +44	Business service instantiated with new instead of dependency injection	Bind service in container and inject via constructor/ __invoke
app/Http/Controllers/Ivr/HistoricalReportsSyncController.php	25, 49, 62, 75, 88, 101, 114, 127, 140, 153, 166, 179 +44	Business service instantiated with new instead of dependency injection	Bind service in container and inject via constructor/ __invoke
app/Http/Controllers/Ivr/HistoricalReportsUpdateController.php	25, 49, 62, 75, 88, 101, 114, 127, 140, 153, 166, 179 +44	Business service instantiated with new instead of dependency injection	Bind service in container and inject via constructor/ __invoke
app/Http/Controllers/Ivr/LiveMonitoringDestroyController.php	25, 49, 62, 75, 88, 101, 114, 127, 140, 153, 166, 179 +44	Business service instantiated with new instead of dependency injection	Bind service in container and inject via constructor/ __invoke
app/Http/Controllers/Ivr/LiveMonitoringExportController.php	25, 49, 62, 75, 88, 101, 114, 127, 140, 153, 166, 179 +44	Business service instantiated with new instead of dependency injection	Bind service in container and inject via constructor/ __invoke
app/Http/Controllers/Ivr/LiveMonitoringImportController.php	25, 49, 62, 75, 88, 101, 114, 127, 140, 153, 166, 179 +44	Business service instantiated with new instead of dependency injection	Bind service in container and inject via constructor/ __invoke
app/Http/Controllers/Ivr/LiveMonitoringIndexController.php	25, 49, 62, 75, 88, 101, 114, 127, 140, 153, 166, 179 +44	Business service instantiated with new instead of dependency injection	Bind service in container and inject via constructor/ __invoke

app/Http/Controllers/Ivr/LiveMonitoringStoreController.php	25, 49, 62, 75, 88, 101, 114, 127, 140, 153, 166, 179 +44	Business service instantiated with new instead of dependency injection	Bind service in container and inject via constructor/ __invoke
app/Http/Controllers/Ivr/LiveMonitoringSyncController.php	25, 49, 62, 75, 88, 101, 114, 127, 140, 153, 166, 179 +44	Business service instantiated with new instead of dependency injection	Bind service in container and inject via constructor/ __invoke
app/Http/Controllers/Ivr/LiveMonitoringUpdateController.php	25, 49, 62, 75, 88, 101, 114, 127, 140, 153, 166, 179 +44	Business service instantiated with new instead of dependency injection	Bind service in container and inject via constructor/ __invoke
app/Http/Controllers/Ivr/PromptLibraryDestroyController.php	25, 49, 62, 75, 88, 101, 114, 127, 140, 153, 166, 179 +44	Business service instantiated with new instead of dependency injection	Bind service in container and inject via constructor/ __invoke
app/Http/Controllers/Ivr/PromptLibraryExportController.php	25, 49, 62, 75, 88, 101, 114, 127, 140, 153, 166, 179 +44	Business service instantiated with new instead of dependency injection	Bind service in container and inject via constructor/ __invoke
app/Http/Controllers/Ivr/PromptLibraryImportController.php	25, 49, 62, 75, 88, 101, 114, 127, 140, 153, 166, 179 +44	Business service instantiated with new instead of dependency injection	Bind service in container and inject via constructor/ __invoke
app/Http/Controllers/Ivr/PromptLibraryIndexController.php	25, 49, 62, 75, 88, 101, 114, 127, 140, 153, 166, 179 +44	Business service instantiated with new instead of dependency injection	Bind service in container and inject via constructor/ __invoke
app/Http/Controllers/Ivr/PromptLibraryStoreController.php	25, 49, 62, 75, 88, 101, 114, 127, 140, 153, 166, 179 +44	Business service instantiated with new instead of dependency injection	Bind service in container and inject via constructor/ __invoke
app/Http/Controllers/Ivr/PromptLibrarySyncController.php	25, 49, 62, 75, 88, 101, 114, 127, 140, 153, 166, 179 +44	Business service instantiated with new instead of dependency injection	Bind service in container and inject via constructor/ __invoke
app/Http/Controllers/Ivr/PromptLibraryUpdateController.php	25, 49, 62, 75, 88, 101, 114, 127, 140, 153, 166, 179 +44	Business service instantiated with new instead of dependency injection	Bind service in container and inject via constructor/ __invoke
app/Http/Controllers/Ivr/QueueManagementDestroyController.php	25, 49, 62, 75, 88, 101, 114, 127, 140, 153, 166, 179 +44	Business service instantiated with new instead of dependency injection	Bind service in container and inject via constructor/ __invoke
app/Http/Controllers/Ivr/QueueManagementExportController.php	25, 49, 62, 75, 88, 101, 114, 127, 140, 153, 166, 179 +44	Business service instantiated with new instead of dependency injection	Bind service in container and inject via constructor/ __invoke
app/Http/Controllers/Ivr/QueueManagementImportController.php	25, 49, 62, 75, 88, 101, 114, 127, 140, 153, 166, 179 +44	Business service instantiated with new instead of dependency injection	Bind service in container and inject via constructor/ __invoke
app/Http/Controllers/Ivr/QueueManagementIndexController.php	25, 49, 62, 75, 88, 101, 114, 127, 140, 153, 166, 179 +44	Business service instantiated with new instead of dependency injection	Bind service in container and inject via constructor/ __invoke
app/Http/Controllers/Ivr/QueueManagementStoreController.php	25, 49, 62, 75, 88, 101, 114,	Business service instantiated with new	Bind service in container and inject via constructor/ __invoke

	127, 140, 153, 166, 179 +44	instead of dependency injection	
<code>app/Http/Controllers/Ivr/QueueManagementSyncController.php</code>	25, 49, 62, 75, 88, 101, 114, 127, 140, 153, 166, 179 +44	Business service instantiated with new instead of dependency injection	Bind service in container and inject via constructor/ __invoke
<code>app/Http/Controllers/Ivr/QueueManagementUpdateController.php</code>	25, 49, 62, 75, 88, 101, 114, 127, 140, 153, 166, 179 +44	Business service instantiated with new instead of dependency injection	Bind service in container and inject via constructor/ __invoke

H5. Missing Service Layer CRITICAL

Benchmark: handlers with inline business logic = **82 IVR controllers** → **High Risk** band (Good <10 · Moderate 10–20 · High Risk >20).

```
// app/Http/Controllers/Ivr/CallAnalyticsStoreController.php:22-30
public function handleStore(Request $request)
{
    // Fat controller - business rules live here
    $q = $request->get("q");
    if ($q) {
        $rows = DB::select("select * from ivr_call_analyticss where name like '%".$q."%' and tenant_id = ".$this->tenantId);
    } else {
        $rows = CallAnalytics::where("tenant_id", $this->tenantId)->get();
    }
}
```

Why it matters here: Filtering, querying, response shaping and tenant handling all live inside the controller, so the same logic cannot be reused from CLI commands, queued jobs or the `ivr-legacy` API without duplication. The "God services" that exist are workflow dumps (`orchestrateXWorkflow1..55`), not a cohesive domain layer, so business intent is spread between fat controllers and fat services.

Recommended approach:

1. Extract one cohesive `CallAnalyticsService` (etc.) per module with intention-revealing methods (`listForTenant` , `record`), replacing the 55 numbered workflow stubs.
2. Reduce each controller to: resolve DTO → call service → return `Inertia::render` / `response()->json` .
3. Cover the extracted services with unit tests before deleting controller logic.

82 files affected.

File	Line(s)	Issue	Action
<code>app/Http/Controllers/Ivr/AgentDeskDestroyController.php</code>	12	Business logic and raw SQL inline in controller (no service tier)	Move workflow logic into a dedicated Service; keep controller thin
<code>app/Http/Controllers/Ivr/AgentDeskExportController.php</code>	12	Business logic and raw SQL inline in controller (no service tier)	Move workflow logic into a dedicated Service; keep controller thin
<code>app/Http/Controllers/Ivr/AgentDeskImportController.php</code>	12	Business logic and raw SQL inline in controller (no service tier)	Move workflow logic into a dedicated Service; keep controller thin

[illegible]

[illegible]

[illegible]

[illegible]

app/Http/Controllers/Ivr/QueueManagementStoreController.php	12	Business logic and raw SQL inline in controller (no service tier)	Move workflow logic into a dedicated Service; keep controller thin
app/Http/Controllers/Ivr/QueueManagementSyncController.php	12	Business logic and raw SQL inline in controller (no service tier)	Move workflow logic into a dedicated Service; keep controller thin
app/Http/Controllers/Ivr/QueueManagementUpdateController.php	12	Business logic and raw SQL inline in controller (no service tier)	Move workflow logic into a dedicated Service; keep controller thin

H6. API Sprawl HIGH

Benchmark: documented & governed endpoints = **0% of 80** **ivr-legacy** routes → **High Risk** band (Good >90% · Moderate 80–90% · High Risk <80%).

```
// routes/generated/ivr_legacy_api.php:5-13
Route::prefix("ivr-legacy")->group(function () {
    Route::match(['get','post'], 'agent-desk/destroy',
App\Http\Controllers\Ivr\AgentDeskDestroyController::class);
    Route::match(['get','post'], 'agent-desk/export',
App\Http\Controllers\Ivr\AgentDeskExportController::class);
    // ...80 routes, each accepting BOTH GET and POST
```

```
// routes/api.php:5-9
require __DIR__.'/generated/ivr_legacy_api.php';
Route::get('/ivr/health-legacy', function () {
    return response()->json(['status' => 'maybe-ok', 'timestamp' => time()]);
});
```

Why it matters here: The same resource is reachable through the Inertia web route (`/ivr/{moduleSlug}`) and through 80 machine-generated `ivr-legacy/*` routes that each accept GET **and** POST, so mutating actions (`destroy` , `store` , `update` , `sync` , `import`) can be triggered by a simple GET. Consumers have no canonical contract and must special-case each verb/route.

Recommended approach:

1. Collapse the 7 per-module actions into RESTful resource routes with correct HTTP verbs (no GET-mutates).
2. Deduplicate the web vs `ivr-legacy` surface behind one versioned API (`/api/v1/...`).
3. Generate the route table from a single source of truth reviewed in PRs, not an auto-synced file.

1 file affected.

File	Line(s)	Issue	Action
routes/generated/ivr_legacy_api.php	7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18 +68	Duplicative endpoint accepting GET+POST for mutating actions	Replace with versioned RESTful resource routes using correct verbs

H7. Missing API Governance HIGH

Benchmark: governance compliance = **0%** — no OpenAPI/Swagger spec, no API versioning, no contract tests anywhere in the repo → **High Risk** band (Good 100% · Moderate 90–99% · High Risk <90%).

The **ivr-legacy** API returns ad-hoc arrays with no schema:

```
// app/Http/Controllers/Ivr/CallAnalyticsStoreController.php:32-33
if ($request->wantsJson()) {
    return response()->json(["data" => $rows, "module" => "CallAnalytics",
    "action" => "Store"]);
}
```

Why it matters here: With 81 endpoints and zero specification, breaking changes ship undetected and integrators reverse-engineer behaviour from responses. There is no **openapi** / **swagger** file, no **/v1** prefix, and no contract test suite to catch drift.

Recommended approach:

1. Author an OpenAPI 3 spec for the consolidated API and publish it (e.g. Scramble or L5-Swagger).
2. Add Spectral API linting and a contract-test job to **.github/workflows**.
3. Version the API (**/api/v1**) and gate merges on spec + contract checks.

81 files affected.

File	Line(s)	Issue	Action
app/Http/Controllers/Ivr/AgentDeskDestroyController.php	34	Undocumented JSON contract with no OpenAPI spec or versioning	Define OpenAPI schema, version the route, add contract tests
app/Http/Controllers/Ivr/AgentDeskExportController.php	34	Undocumented JSON contract with no OpenAPI spec or versioning	Define OpenAPI schema, version the route, add contract tests
app/Http/Controllers/Ivr/AgentDeskImportController.php	34	Undocumented JSON contract with no OpenAPI spec or versioning	Define OpenAPI schema, version the route, add contract tests
app/Http/Controllers/Ivr/AgentDeskIndexController.php	34	Undocumented JSON contract with no OpenAPI spec or versioning	Define OpenAPI schema, version the route, add contract tests
app/Http/Controllers/Ivr/AgentDeskStoreController.php	34	Undocumented JSON contract with no OpenAPI spec or versioning	Define OpenAPI schema, version the route, add contract tests
app/Http/Controllers/Ivr/AgentDeskSyncController.php	34	Undocumented JSON contract with no OpenAPI spec or versioning	Define OpenAPI schema, version the route, add contract tests
app/Http/Controllers/Ivr/AgentDeskUpdateController.php	34	Undocumented JSON contract with no OpenAPI spec or versioning	Define OpenAPI schema, version the route, add contract tests
app/Http/Controllers/Ivr/BusinessHoursDestroyController.php	34	Undocumented JSON contract with no OpenAPI spec or versioning	Define OpenAPI schema, version the route, add contract tests
app/Http/Controllers/Ivr/BusinessHoursExportController.php	34	Undocumented JSON contract with no OpenAPI spec or versioning	Define OpenAPI schema, version the route, add contract tests
app/Http/Controllers/Ivr/BusinessHoursImportController.php	34	Undocumented JSON contract with no OpenAPI spec or versioning	Define OpenAPI schema, version the route, add contract tests
app/Http/Controllers/Ivr/BusinessHoursIndexController.php	34	Undocumented JSON contract with no OpenAPI spec or versioning	Define OpenAPI schema, version the route, add contract tests

[illegible]

[illegible]

app/Http/Controllers/Ivr/DidInventoryStoreController.php	34	Undocumented JSON contract with no OpenAPI spec or versioning	Define OpenAPI schema, version the route, add contract tests
app/Http/Controllers/Ivr/DidInventorySyncController.php	34	Undocumented JSON contract with no OpenAPI spec or versioning	Define OpenAPI schema, version the route, add contract tests
app/Http/Controllers/Ivr/DidInventoryUpdateController.php	34	Undocumented JSON contract with no OpenAPI spec or versioning	Define OpenAPI schema, version the route, add contract tests
app/Http/Controllers/Ivr/HistoricalReportsDestroyController.php	34	Undocumented JSON contract with no OpenAPI spec or versioning	Define OpenAPI schema, version the route, add contract tests
app/Http/Controllers/Ivr/HistoricalReportsExportController.php	34	Undocumented JSON contract with no OpenAPI spec or versioning	Define OpenAPI schema, version the route, add contract tests
app/Http/Controllers/Ivr/HistoricalReportsImportController.php	34	Undocumented JSON contract with no OpenAPI spec or versioning	Define OpenAPI schema, version the route, add contract tests
app/Http/Controllers/Ivr/HistoricalReportsIndexController.php	34	Undocumented JSON contract with no OpenAPI spec or versioning	Define OpenAPI schema, version the route, add contract tests
app/Http/Controllers/Ivr/HistoricalReportsStoreController.php	34	Undocumented JSON contract with no OpenAPI spec or versioning	Define OpenAPI schema, version the route, add contract tests
app/Http/Controllers/Ivr/HistoricalReportsSyncController.php	34	Undocumented JSON contract with no OpenAPI spec or versioning	Define OpenAPI schema, version the route, add contract tests
app/Http/Controllers/Ivr/HistoricalReportsUpdateController.php	34	Undocumented JSON contract with no OpenAPI spec or versioning	Define OpenAPI schema, version the route, add contract tests
app/Http/Controllers/Ivr/IvrHubController.php	36	Undocumented JSON contract with no OpenAPI spec or versioning	Define OpenAPI schema, version the route, add contract tests
app/Http/Controllers/Ivr/LiveMonitoringDestroyController.php	34	Undocumented JSON contract with no OpenAPI spec or versioning	Define OpenAPI schema, version the route, add contract tests
app/Http/Controllers/Ivr/LiveMonitoringExportController.php	34	Undocumented JSON contract with no OpenAPI spec or versioning	Define OpenAPI schema, version the route, add contract tests
app/Http/Controllers/Ivr/LiveMonitoringImportController.php	34	Undocumented JSON contract with no OpenAPI spec or versioning	Define OpenAPI schema, version the route, add contract tests
app/Http/Controllers/Ivr/LiveMonitoringIndexController.php	34	Undocumented JSON contract with no OpenAPI spec or versioning	Define OpenAPI schema, version the route, add contract tests
app/Http/Controllers/Ivr/LiveMonitoringStoreController.php	34	Undocumented JSON contract with no OpenAPI spec or versioning	Define OpenAPI schema, version the route, add contract tests
app/Http/Controllers/Ivr/LiveMonitoringSyncController.php	34	Undocumented JSON contract with no OpenAPI spec or versioning	Define OpenAPI schema, version the route, add contract tests
app/Http/Controllers/Ivr/LiveMonitoringUpdateController.php	34	Undocumented JSON contract with no OpenAPI spec or versioning	Define OpenAPI schema, version the route, add contract tests
app/Http/Controllers/Ivr/PromptLibraryDestroyController.php	34	Undocumented JSON contract with no OpenAPI spec or versioning	Define OpenAPI schema, version the route, add contract tests

<code>app/Http/Controllers/Ivr/PromptLibraryExportController.php</code>	34	Undocumented JSON contract with no OpenAPI spec or versioning	Define OpenAPI schema, version the route, add contract tests
<code>app/Http/Controllers/Ivr/PromptLibraryImportController.php</code>	34	Undocumented JSON contract with no OpenAPI spec or versioning	Define OpenAPI schema, version the route, add contract tests
<code>app/Http/Controllers/Ivr/PromptLibraryIndexController.php</code>	34	Undocumented JSON contract with no OpenAPI spec or versioning	Define OpenAPI schema, version the route, add contract tests
<code>app/Http/Controllers/Ivr/PromptLibraryStoreController.php</code>	34	Undocumented JSON contract with no OpenAPI spec or versioning	Define OpenAPI schema, version the route, add contract tests
<code>app/Http/Controllers/Ivr/PromptLibrarySyncController.php</code>	34	Undocumented JSON contract with no OpenAPI spec or versioning	Define OpenAPI schema, version the route, add contract tests
<code>app/Http/Controllers/Ivr/PromptLibraryUpdateController.php</code>	34	Undocumented JSON contract with no OpenAPI spec or versioning	Define OpenAPI schema, version the route, add contract tests
<code>app/Http/Controllers/Ivr/QueueManagementDestroyController.php</code>	34	Undocumented JSON contract with no OpenAPI spec or versioning	Define OpenAPI schema, version the route, add contract tests
<code>app/Http/Controllers/Ivr/QueueManagementExportController.php</code>	34	Undocumented JSON contract with no OpenAPI spec or versioning	Define OpenAPI schema, version the route, add contract tests
<code>app/Http/Controllers/Ivr/QueueManagementImportController.php</code>	34	Undocumented JSON contract with no OpenAPI spec or versioning	Define OpenAPI schema, version the route, add contract tests
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<code>app/Http/Controllers/Ivr/QueueManagementSyncController.php</code>	34	Undocumented JSON contract with no OpenAPI spec or versioning	Define OpenAPI schema, version the route, add contract tests
<code>app/Http/Controllers/Ivr/QueueManagementUpdateController.php</code>	34	Undocumented JSON contract with no OpenAPI spec or versioning	Define OpenAPI schema, version the route, add contract tests

H8. Weak Application Architecture Pattern HIGH

Benchmark: modules following declared MVC/layered architecture = **<20%** (core CRM compliant; entire IVR subsystem non-compliant) → **High Risk** band (Good >80% · Moderate 50–80% · High Risk <50%).

```
// app/Models/Ivr/CallAnalytics.php:22-24
protected $guarded = []; // legacy - mass assignment wide open
// ...domain logic + raw SQL living inside the model:
public function legacyComputedField1()
{
    return DB::select("select count(*) as c from ivr_call_analyticss where
tenant_id = ?", [$this->tenant_id ?? 1]);
}
```

Why it matters here: The declared MVC separation is honoured by the CRM controllers but abandoned in IVR: HTTP handling, business rules and raw SQL sit in controllers, while models carry 35 query-executing accessor methods. A change to one IVR module's storage cascades into controller, model and service edits, and new developers cannot predict where logic lives.

Recommended approach:

1. Enforce layer boundaries: HTTP (controllers) → domain (services) → data (repositories); models hold relations/casts only.
2. Move `legacyComputedField*` query methods out of `App\Models\Ivr\*` into repositories.
3. Add architecture tests (Pest arch) asserting models contain no `DB::` and controllers contain no business rules.

12 files affected.

File	Line(s)	Issue	Action
<code>app/Models/Ivr/AgentDesk.php</code>	25, 31, 37, 43, 49, 55, 61, 67, 73, 79, 85, 91 +23	Domain queries and logic embedded in Eloquent model (layer violation)	Move query methods to repository; keep model to relations/casts
<code>app/Models/Ivr/BusinessHours.php</code>	25, 31, 37, 43, 49, 55, 61, 67, 73, 79, 85, 91 +23	Domain queries and logic embedded in Eloquent model (layer violation)	Move query methods to repository; keep model to relations/casts
<code>app/Models/Ivr/CallAnalytics.php</code>	25, 31, 37, 43, 49, 55, 61, 67, 73, 79, 85, 91 +23	Domain queries and logic embedded in Eloquent model (layer violation)	Move query methods to repository; keep model to relations/casts
<code>app/Models/Ivr/CallFlow.php</code>	25, 31, 37, 43, 49, 55, 61, 67, 73, 79, 85, 91 +23	Domain queries and logic embedded in Eloquent model (layer violation)	Move query methods to repository; keep model to relations/casts
<code>app/Models/Ivr/CallRecording.php</code>	25, 31, 37, 43, 49, 55, 61, 67, 73, 79, 85, 91 +23	Domain queries and logic embedded in Eloquent model (layer violation)	Move query methods to repository; keep model to relations/casts
<code>app/Models/Ivr/CallRouting.php</code>	25, 31, 37, 43, 49, 55, 61, 67, 73, 79, 85, 91 +23	Domain queries and logic embedded in Eloquent model (layer violation)	Move query methods to repository; keep model to relations/casts
<code>app/Models/Ivr/CustomerProfile.php</code>	25, 31, 37, 43, 49, 55, 61, 67, 73, 79, 85, 91 +23	Domain queries and logic embedded in Eloquent model (layer violation)	Move query methods to repository; keep model to relations/casts
<code>app/Models/Ivr/DidInventory.php</code>	25, 31, 37, 43, 49, 55, 61, 67, 73, 79, 85, 91 +23	Domain queries and logic embedded in Eloquent model (layer violation)	Move query methods to repository; keep model to relations/casts
<code>app/Models/Ivr/HistoricalReports.php</code>	25, 31, 37, 43, 49, 55, 61, 67, 73, 79, 85, 91 +23	Domain queries and logic embedded in Eloquent model (layer violation)	Move query methods to repository; keep model to relations/casts
<code>app/Models/Ivr/LiveMonitoring.php</code>	25, 31, 37, 43, 49, 55, 61, 67, 73, 79, 85, 91 +23	Domain queries and logic embedded in Eloquent model (layer violation)	Move query methods to repository; keep model to relations/casts
<code>app/Models/Ivr/PromptLibrary.php</code>	25, 31, 37, 43, 49, 55, 61, 67, 73, 79, 85, 91 +23	Domain queries and logic embedded in Eloquent model (layer violation)	Move query methods to repository; keep model to relations/casts
<code>app/Models/Ivr/QueueManagement.php</code>	25, 31, 37, 43, 49, 55, 61, 67, 73, 79, 85, 91 +23	Domain queries and logic embedded in Eloquent model (layer violation)	Move query methods to repository; keep model to relations/casts

H9. Missing Module / Package Inventory MEDIUM

Benchmark: circular dependency count = 0, but 17 dead files (5 `LegacyIvr*` helpers + 12 `Repositories/Legacy`) are never referenced from any entry point → **Moderate** band (Good 0 · Moderate 1–3 · High Risk >3 for circular deps; rated Moderate on dead-module evidence).

```
# grep for references outside their own folder returns nothing:
grep -r "LegacyIvr" app --include=*.php | grep -v app/Legacy/Helpers →
(empty)
grep -r "Repository" app --include=*.php | grep -v app/Repositories →
(empty)
```

Why it matters here: ~5,600 lines across 17 files (2,835 lines of static helpers + 12 repositories) are unreachable dead code that still ships, is scanned by tools, and misleads developers into thinking a data layer is in use. No circular dependencies were detected, so the risk is maintenance drag and duplicate-logic confusion rather than initialization order.

Recommended approach:

1. Confirm zero runtime references (static analysis) and delete or archive the unused `LegacyIvr*` helpers.
2. Either wire the `Repositories/Legacy` classes into the data layer (see H3) or remove them.
3. Document each surviving module's public API in a short `MODULES.md`.

5 files affected.

File	Line(s)	Issue	Action
<code>app/Legacy/Helpers/LegacyIvrArray.php</code>	5	Unreferenced dead helper class (no call sites in the codebase)	Delete/archive after confirming zero references
<code>app/Legacy/Helpers/LegacyIvrCrypto.php</code>	5	Unreferenced dead helper class (no call sites in the codebase)	Delete/archive after confirming zero references
<code>app/Legacy/Helpers/LegacyIvrDate.php</code>	5	Unreferenced dead helper class (no call sites in the codebase)	Delete/archive after confirming zero references
<code>app/Legacy/Helpers/LegacyIvrMath.php</code>	5	Unreferenced dead helper class (no call sites in the codebase)	Delete/archive after confirming zero references
<code>app/Legacy/Helpers/LegacyIvrString.php</code>	5	Unreferenced dead helper class (no call sites in the codebase)	Delete/archive after confirming zero references

H11. Middleware & Filter Weakness HIGH

Benchmark: required middleware present & correctly ordered = **partial** — auth+throttle present, but no security-headers, no request-ID/audit logging, no explicit CORS policy → **Moderate** band (Good 100% · Moderate 80–99% · High Risk <80%).

```
// bootstrap/app.php:17-25
->withMiddleware(function (Middleware $middleware) {
    $middleware->redirectGuestsTo(fn () => route('login'));
    $middleware->web(\App\Http\Middleware\HandleInertiaRequests::class);
    $middleware->throttleApi(); // throttle present...
    // ...no security headers, no correlation-id logging, no CORS config
});
```

```
// config/ivr_legacy.php:14 - auth-bypass switch consumed by legacy code
'bypass_auth_for_internal_ips' => ['127.0.0.1', '10.0.0.0'],
```

Why it matters here: There is no `helmet`-equivalent security-headers middleware, no structured request logging with correlation IDs (making the swallowed-exception problem in H18 impossible to trace), and no `config/cors.php` so CORS falls to

framework defaults. Combined with an IP-based auth-bypass flag, the pipeline cannot enforce a consistent security posture.

Recommended approach:

1. Add a security-headers middleware (CSP, X-Frame-Options, HSTS) to the web/api groups.
2. Add a correlation-ID + structured request-logging middleware early in the stack.
3. Publish an explicit `config/cors.php` with an allow-list; remove `bypass_auth_for_internal_ips`.

1 file affected.

File	Line(s)	Issue	Action
<code>bootstrap/app.php</code>	17	Missing security-headers / request-logging middleware; IP auth-bypass	Add security-headers + correlation-id logging; remove auth bypass

H12. Authentication & Authorization Weakness CRITICAL

Benchmark: protected routes guarded ≈ 26% (CRM web routes guarded; 80 `ivr-legacy` API routes have zero `auth` middleware) + object-level IDOR; password hashing = `bcrypt (strong)` → **High Risk** band (broken access control drives the verdict despite strong hashing).

```
// routes/generated/ivr_legacy_api.php – no ->middleware('auth') anywhere (grep
count = 0)
Route::prefix("ivr-legacy")->group(function () {
    Route::match(['get','post'], 'call-recording/export',
App\Http\Controllers\Ivr\CallRecordingExportController::class);
```

```
// app/Http/Controllers/Ivr/CallAnalyticsStoreController.php:14-15
// AUTH-NOTE: some endpoints intentionally skip policies (2014 regression)
private $tenantId = 1; // hard-coded tenant – multi-tenant broken
```

Why it matters here: 80 state-changing endpoints (`store` , `update` , `destroy` , `import` , `sync` , `export`) are reachable **unauthenticated**, and every IVR query is scoped by a hard-coded `tenant_id = 1` instead of the caller's tenant — so any user can read/mutate any tenant's data (classic IDOR / OWASP #1 Broken Access Control). Password hashing itself is sound (`Hash::make` , `bcrypt` in `User::setPasswordAttribute`).

Recommended approach:

1. Wrap `routes/generated/ivr_legacy_api.php` in `Route::middleware(['auth:sanctum'])` (or remove the surface entirely).
2. Replace hard-coded `$tenantId = 1` with the authenticated user's account/tenant via `IvrAccountContext::fromRequest` , and add Policy object-level checks in services.
3. Delete the `bypass_auth_for_internal_ips` and "skip policies" behaviour; add tests asserting cross-tenant access is denied.

80 files affected.

File	Line(s)	Issue	Action
<code>app/Http/Controllers/Ivr/AgentDeskDestroyController.php</code>	15	Hard-coded tenant id breaks multi-tenant isolation (IDOR)	Derive tenant from authenticated user; add object-level authorization

[illegible]

[illegible]

[illegible]

app/Http/Controllers/Ivr/QueueManagementUpdateController.php	15	Hard-coded tenant id breaks multi-tenant isolation (IDOR)	Derive tenant from authenticated user; add object-level authorization
--	----	---	---

H13. Backend Security Vulnerabilities **CRITICAL**

Benchmark: injection + hardcoded-secret patterns = >3 (SQLi + mass-assignment + debug-on + committed secrets) → **High Risk** band (Good 0 each · Moderate 1–3 total · High Risk >3).

```
// app/Repositories/Legacy/CallAnalyticsRepository.php:14 – SQL injection
$sql .= " AND name LIKE '%" . $filter . "%'"; // unbound user input
```

```
// app/Models/Ivr/CallAnalytics.php:23 – mass assignment wide open
protected $guarded = [];
```

```
# .env.example: ships debug-on defaults
APP_ENV=local
APP_DEBUG=true
```

Why it matters here: User-controlled `$q / $filter` values are concatenated straight into `DB::select(...)` in controllers and repositories (injectable), `$guarded = []` lets any request field write any column, and `APP_DEBUG=true` (plus H18 leaking `$e->getMessage()`) exposes stack traces. Together these cover three of the top OWASP categories in one subsystem.

Recommended approach:

1. Replace all string-concatenated SQL with bound parameters / query-builder `where(...)` (ties to H3).
2. Replace `$guarded = []` with explicit `$fillable` allow-lists on every `App\Models\Ivr\*` model.
3. Ship `APP_DEBUG=false` for non-local envs and stop returning exception messages to clients (H18).

12 files affected.

File	Line(s)	Issue	Action
app/Models/Ivr/AgentDesk.php	14	Mass assignment fully open (guarded = []) enabling column injection	Replace with explicit \$fillable allow-list per model
app/Models/Ivr/BusinessHours.php	14	Mass assignment fully open (guarded = []) enabling column injection	Replace with explicit \$fillable allow-list per model
app/Models/Ivr/CallAnalytics.php	14	Mass assignment fully open (guarded = []) enabling column injection	Replace with explicit \$fillable allow-list per model
app/Models/Ivr/CallFlow.php	14	Mass assignment fully open (guarded = []) enabling column injection	Replace with explicit \$fillable allow-list per model
app/Models/Ivr/CallRecording.php	14	Mass assignment fully open (guarded = []) enabling column injection	Replace with explicit \$fillable allow-list per model
app/Models/Ivr/CallRouting.php	14	Mass assignment fully open (guarded = []) enabling column injection	Replace with explicit \$fillable allow-list per model
app/Models/Ivr/CustomerProfile.php	14	Mass assignment fully open (guarded = []) enabling column injection	Replace with explicit \$fillable allow-list per model
app/Models/Ivr/DidInventory.php	14	Mass assignment fully open (guarded = []) enabling column injection	Replace with explicit \$fillable allow-list per model

app/Models/Ivr/HistoricalReports.php	14	Mass assignment fully open (guarded = []) enabling column injection	Replace with explicit \$fillable allow-list per model
app/Models/Ivr/LiveMonitoring.php	14	Mass assignment fully open (guarded = []) enabling column injection	Replace with explicit \$fillable allow-list per model
app/Models/Ivr/PromptLibrary.php	14	Mass assignment fully open (guarded = []) enabling column injection	Replace with explicit \$fillable allow-list per model
app/Models/Ivr/QueueManagement.php	14	Mass assignment fully open (guarded = []) enabling column injection	Replace with explicit \$fillable allow-list per model

H14. Performance & Caching Gaps HIGH

Benchmark: N+1 / blocking patterns = **420 N+1 accessor methods** + **540 synchronous sleep(1)** + **0 caching** → **High Risk** band (Good 0 · Moderate 1–5 · High Risk >5).

```
// app/Models/Ivr/CallAnalytics.php:26-30 – 35 such accessors × 12 models = 420
public function legacyComputedField1()
{
    // N+1 friendly accessor – called from blade/react randomly
    return DB::select("select count(*) as c from ivr_call_analyticss where
tenant_id = ?", [$this->tenant_id ?? 1]);
}
```

```
// app/Legacy/Services/CallAnalyticsGodService.php:16-18 – 45 methods × 12
services = 540
extract($payload);
sleep(1); // blocking synchronous remote sync
self::$sharedRuntimeCache[$tenant_id ?? 1] = DB::table("ivr_call_analyticss")-
>insertGetId((array) $payload);
```

Why it matters here: Each God-service workflow blocks the PHP-FPM worker for a full second (sleep(1)), so a handful of concurrent calls exhaust the worker pool; the 420 per-model count accessors run one query each when iterated; and Cache:: / cache() appears **0 times** in app/, so identical hot queries hit the database on every request with no Cache-Control headers on read endpoints.

Recommended approach:

1. Remove sleep(1) ; move any real remote sync to a queued job (ShouldQueue).
2. Replace legacyComputedField* per-row queries with a single aggregated query / eager load.
3. Introduce Redis caching for hot read queries and add Cache-Control headers to GET endpoints.

12 files affected.

File	Line(s)	Issue	Action
app/Legacy/Services/AgentDeskGodService.php	16, 24, 32, 40, 48, 56, 64, 72, 80, 88, 96, 104 +33	Synchronous blocking sleep on the request path (worker starvation)	Remove sleep; offload remote sync to a queued job
app/Legacy/Services/BusinessHoursGodService.php	16, 24, 32, 40, 48, 56, 64, 72, 80, 88, 96, 104	Synchronous blocking sleep on the request path (worker	Remove sleep; offload remote sync to a queued

	+33	starvation)	job
app/Legacy/Services/CallAnalyticsGodService.php	16, 24, 32, 40, 48, 56, 64, 72, 80, 88, 96, 104 +33	Synchronous blocking sleep on the request path (worker starvation)	Remove sleep; offload remote sync to a queued job
app/Legacy/Services/CallFlowGodService.php	16, 24, 32, 40, 48, 56, 64, 72, 80, 88, 96, 104 +33	Synchronous blocking sleep on the request path (worker starvation)	Remove sleep; offload remote sync to a queued job
app/Legacy/Services/CallRecordingGodService.php	16, 24, 32, 40, 48, 56, 64, 72, 80, 88, 96, 104 +33	Synchronous blocking sleep on the request path (worker starvation)	Remove sleep; offload remote sync to a queued job
app/Legacy/Services/CallRoutingGodService.php	16, 24, 32, 40, 48, 56, 64, 72, 80, 88, 96, 104 +33	Synchronous blocking sleep on the request path (worker starvation)	Remove sleep; offload remote sync to a queued job
app/Legacy/Services/CustomerProfileGodService.php	16, 24, 32, 40, 48, 56, 64, 72, 80, 88, 96, 104 +33	Synchronous blocking sleep on the request path (worker starvation)	Remove sleep; offload remote sync to a queued job
app/Legacy/Services/DidInventoryGodService.php	16, 24, 32, 40, 48, 56, 64, 72, 80, 88, 96, 104 +33	Synchronous blocking sleep on the request path (worker starvation)	Remove sleep; offload remote sync to a queued job
app/Legacy/Services/HistoricalReportsGodService.php	16, 24, 32, 40, 48, 56, 64, 72, 80, 88, 96, 104 +33	Synchronous blocking sleep on the request path (worker starvation)	Remove sleep; offload remote sync to a queued job
app/Legacy/Services/LiveMonitoringGodService.php	16, 24, 32, 40, 48, 56, 64, 72, 80, 88, 96, 104 +33	Synchronous blocking sleep on the request path (worker starvation)	Remove sleep; offload remote sync to a queued job
app/Legacy/Services/PromptLibraryGodService.php	16, 24, 32, 40, 48, 56, 64, 72, 80, 88, 96, 104 +33	Synchronous blocking sleep on the request path (worker starvation)	Remove sleep; offload remote sync to a queued job
app/Legacy/Services/QueueManagementGodService.php	16, 24, 32, 40, 48, 56, 64, 72, 80, 88, 96, 104 +33	Synchronous blocking sleep on the request path (worker starvation)	Remove sleep; offload remote sync to a queued job

H16. Secrets & Configuration in Source CRITICAL

Benchmark: hardcoded secrets / committed `.env` = 15+ (`.env` is correctly git-ignored, but secrets are hardcoded in tracked source) → **High Risk** band (Good 0 · Moderate 1–2 · High Risk >2).

```
// config/ivr_legacy.php:11-24
'master_api_key' => 'IVR-MASTER-KEY-DO-NOT-COMMIT-2013',
'crm' => ['salesforce' => [
    'client_secret' => 'hardcoded_sf_secret_2015',
    'password'      => 'PlainTextPassword!',
]],
```

```
// app/Legacy/Services/CallAnalyticsGodService.php:11  (×12 services, distinct keys)
private $apiKey = "LEGACY_IVR_KEY_2082"; // hard-coded secret
```

Why it matters here: A tracked config file commits a master API key, a Salesforce client secret and a plaintext integration password, and 12 services each embed their own API key — anyone with repository read access obtains production credentials and can move

laterally into Salesforce. `.gitignore` does exclude `.env`, which limits blast radius but does not help these hardcoded values.

Recommended approach:

1. Move every literal in `config/ivr_legacy.php` and the 12 service `$apiKey` values to `env()`-backed config or a secrets manager (Vault/AWS Secrets Manager).
2. Rotate all exposed keys/passwords immediately — they must be treated as compromised.
3. Add a secret-scanning step (gitleaks/trufflehog) to CI to block future commits.

12 files affected.

File	Line(s)	Issue	Action
<code>app/Legacy/Services/AgentDeskGodService.php</code>	11	Credential/API key hardcoded in tracked source	Move to env/secrets manager and rotate the exposed value
<code>app/Legacy/Services/BusinessHoursGodService.php</code>	11	Credential/API key hardcoded in tracked source	Move to env/secrets manager and rotate the exposed value
<code>app/Legacy/Services/CallAnalyticsGodService.php</code>	11	Credential/API key hardcoded in tracked source	Move to env/secrets manager and rotate the exposed value
<code>app/Legacy/Services/CallFlowGodService.php</code>	11	Credential/API key hardcoded in tracked source	Move to env/secrets manager and rotate the exposed value
<code>app/Legacy/Services/CallRecordingGodService.php</code>	11	Credential/API key hardcoded in tracked source	Move to env/secrets manager and rotate the exposed value
<code>app/Legacy/Services/CallRoutingGodService.php</code>	11	Credential/API key hardcoded in tracked source	Move to env/secrets manager and rotate the exposed value
<code>app/Legacy/Services/CustomerProfileGodService.php</code>	11	Credential/API key hardcoded in tracked source	Move to env/secrets manager and rotate the exposed value
<code>app/Legacy/Services/DidInventoryGodService.php</code>	11	Credential/API key hardcoded in tracked source	Move to env/secrets manager and rotate the exposed value
<code>app/Legacy/Services/HistoricalReportsGodService.php</code>	11	Credential/API key hardcoded in tracked source	Move to env/secrets manager and rotate the exposed value
<code>app/Legacy/Services/LiveMonitoringGodService.php</code>	11	Credential/API key hardcoded in tracked source	Move to env/secrets manager and rotate the exposed value
<code>app/Legacy/Services/PromptLibraryGodService.php</code>	11	Credential/API key hardcoded in tracked source	Move to env/secrets manager and rotate the exposed value
<code>app/Legacy/Services/QueueManagementGodService.php</code>	11	Credential/API key hardcoded in tracked source	Move to env/secrets manager and rotate the exposed value

H17. Backend Code Quality Issues MEDIUM

Benchmark: linter in CI = **Yes** (coding-standards + PHPStan), but PHPStan pinned to **level 1** with pervasive duplication/dead code → **Moderate** band (Good both · Moderate one gap · High Risk both bad).

```
# .github/workflows/static-analysis.yml – runs, but...
# phpstan.neon:9 → level: 1 (of a possible 9)
```

```
// app/Legacy/Services/CallAnalyticsGodService.php – 55 near-identical numbered methods
public function orchestrateCallAnalyticsWorkflow2($payload) {
    extract($payload); sleep(1); /* ...identical... */ }
```

```
public function orchestrateCallAnalyticsWorkflow3($payload) {
    extract($payload); sleep(1); /* ...identical... */ }
```

Why it matters here: CI does run PHPStan and a code-style workflow (a real strength), but level 1 catches almost nothing, so the ~4,400 copy-pasted workflow/endpoint methods, 2,835 lines of dead static helpers and the models' 420 duplicated accessors all pass unchallenged. Individual functions are short (low per-function cyclomatic complexity), so the quality problem is duplication and scale, not branching depth. TODO/FIXME density is 0.

Recommended approach:

1. Raise `phpstan.neon` level incrementally (target ≥ 5) after the DTO work removes dynamic vars.
2. Add a duplication/complexity gate (e.g. PHPMD, `phpcpd`) to CI and fail on cloned blocks.
3. Delete generated duplication by collapsing numbered methods into single parameterised methods.

12 files affected.

File	Line(s)	Issue	Action
<code>app/Legacy/Services/AgentDeskGodService.php</code>	13, 21, 29, 37, 45, 53, 61, 69, 77, 85, 93, 101 +33	Massive copy-paste duplication (numbered near-identical methods)	Collapse into a single parameterised method; raise PHPStan level
<code>app/Legacy/Services/BusinessHoursGodService.php</code>	13, 21, 29, 37, 45, 53, 61, 69, 77, 85, 93, 101 +33	Massive copy-paste duplication (numbered near-identical methods)	Collapse into a single parameterised method; raise PHPStan level
<code>app/Legacy/Services/CallAnalyticsGodService.php</code>	13, 21, 29, 37, 45, 53, 61, 69, 77, 85, 93, 101 +33	Massive copy-paste duplication (numbered near-identical methods)	Collapse into a single parameterised method; raise PHPStan level
<code>app/Legacy/Services/CallFlowGodService.php</code>	13, 21, 29, 37, 45, 53, 61, 69, 77, 85, 93, 101 +33	Massive copy-paste duplication (numbered near-identical methods)	Collapse into a single parameterised method; raise PHPStan level
<code>app/Legacy/Services/CallRecordingGodService.php</code>	13, 21, 29, 37, 45, 53, 61, 69, 77, 85, 93, 101 +33	Massive copy-paste duplication (numbered near-identical methods)	Collapse into a single parameterised method; raise PHPStan level
<code>app/Legacy/Services/CallRoutingGodService.php</code>	13, 21, 29, 37, 45, 53, 61, 69, 77, 85, 93, 101 +33	Massive copy-paste duplication (numbered near-identical methods)	Collapse into a single parameterised method; raise PHPStan level
<code>app/Legacy/Services/CustomerProfileGodService.php</code>	13, 21, 29, 37, 45, 53, 61, 69, 77, 85, 93, 101 +33	Massive copy-paste duplication (numbered near-identical methods)	Collapse into a single parameterised method; raise PHPStan level
<code>app/Legacy/Services/DidInventoryGodService.php</code>	13, 21, 29, 37, 45, 53, 61, 69, 77, 85, 93, 101 +33	Massive copy-paste duplication (numbered near-identical methods)	Collapse into a single parameterised method; raise PHPStan level
<code>app/Legacy/Services/HistoricalReportsGodService.php</code>	13, 21, 29, 37, 45, 53, 61, 69, 77, 85, 93, 101 +33	Massive copy-paste duplication (numbered near-identical methods)	Collapse into a single parameterised method; raise PHPStan level
<code>app/Legacy/Services/LiveMonitoringGodService.php</code>	13, 21, 29, 37, 45, 53, 61, 69, 77, 85, 93, 101 +33	Massive copy-paste duplication (numbered near-identical methods)	Collapse into a single parameterised method; raise PHPStan level
<code>app/Legacy/Services/PromptLibraryGodService.php</code>	13, 21, 29, 37, 45, 53, 61, 69, 77, 85, 93, 101 +33	Massive copy-paste duplication (numbered near-identical methods)	Collapse into a single parameterised method; raise PHPStan level
<code>app/Legacy/Services/QueueManagementGodService.php</code>	13, 21, 29, 37, 45, 53, 61, 69, 77, 85, 93, 101 +33	Massive copy-paste duplication (numbered near-identical methods)	Collapse into a single parameterised method; raise PHPStan level

H18. Swallowed Exceptions & Silent Failure (additional) HIGH

Benchmark: silent/leaky `catch` blocks = ~4,400 → **High Risk** band (Good 0 · Moderate 1–20 · High Risk >20).

```
// app/Http/Controllers/Ivr/CallAnalyticsStoreController.php:50-54 (repeated
~4,400x)
} catch (\Throwable $e) {
    return ["ok" => false, "err" => $e->getMessage()]; // swallowed stack
traces
}
```

Why it matters here: Every `LegacyEndpoint*` wraps its work in a blanket `catch (\Throwable)` that returns a normal HTTP 200 payload — failures never surface to logs, metrics or the client's error handling, so data-loss bugs are invisible in production. Simultaneously it leaks the raw exception message (DB errors, file paths) to the caller, compounding H13.

Recommended approach:

1. Remove blanket catches; let the framework exception handler log and return sanitized responses.
2. Where a catch is legitimately needed, log with a correlation ID (H11) and rethrow or return a proper error status — never a success shape.
3. Add a lint rule forbidding empty/ `getMessage()` -returning `catch (\Throwable)` blocks.

81 files affected.

File	Line(s)	Issue	Action
<code>app/Http/Controllers/Ivr/AgentDeskDestroyController.php</code>	52, 65, 78, 91, 104, 117, 130, 143, 156, 169, 182, 195 +43	Blanket catch swallows failures and leaks exception messages	Remove catch or log+rethrow; never return a success-shaped payload
<code>app/Http/Controllers/Ivr/AgentDeskExportController.php</code>	52, 65, 78, 91, 104, 117, 130, 143, 156, 169, 182, 195 +43	Blanket catch swallows failures and leaks exception messages	Remove catch or log+rethrow; never return a success-shaped payload
<code>app/Http/Controllers/Ivr/AgentDeskImportController.php</code>	52, 65, 78, 91, 104, 117, 130, 143, 156, 169, 182, 195 +43	Blanket catch swallows failures and leaks exception messages	Remove catch or log+rethrow; never return a success-shaped payload
<code>app/Http/Controllers/Ivr/AgentDeskIndexController.php</code>	52, 65, 78, 91, 104, 117, 130, 143, 156, 169, 182, 195 +43	Blanket catch swallows failures and leaks exception messages	Remove catch or log+rethrow; never return a success-shaped payload
<code>app/Http/Controllers/Ivr/AgentDeskStoreController.php</code>	52, 65, 78, 91, 104, 117, 130, 143, 156, 169, 182, 195 +43	Blanket catch swallows failures and leaks exception messages	Remove catch or log+rethrow; never return a success-shaped payload
<code>app/Http/Controllers/Ivr/AgentDeskSyncController.php</code>	52, 65, 78, 91, 104, 117, 130, 143, 156, 169, 182, 195 +43	Blanket catch swallows failures and leaks exception messages	Remove catch or log+rethrow; never return a success-shaped payload
<code>app/Http/Controllers/Ivr/AgentDeskUpdateController.php</code>	52, 65, 78, 91, 104, 117, 130, 143, 156, 169, 182, 195 +43	Blanket catch swallows failures and leaks exception messages	Remove catch or log+rethrow; never return a success-shaped payload
<code>app/Http/Controllers/Ivr/BusinessHoursDestroyController.php</code>	52, 65, 78, 91, 104, 117, 130, 143, 156, 169, 182, 195 +43	Blanket catch swallows failures and leaks exception messages	Remove catch or log+rethrow; never return a success-shaped payload

app/Http/Controllers/Ivr/BusinessHoursExportController.php	52, 65, 78, 91, 104, 117, 130, 143, 156, 169, 182, 195 +43	Blanket catch swallows failures and leaks exception messages	Remove catch or log+rethrow; never return a success-shaped payload
app/Http/Controllers/Ivr/BusinessHoursImportController.php	52, 65, 78, 91, 104, 117, 130, 143, 156, 169, 182, 195 +43	Blanket catch swallows failures and leaks exception messages	Remove catch or log+rethrow; never return a success-shaped payload
app/Http/Controllers/Ivr/BusinessHoursIndexController.php	52, 65, 78, 91, 104, 117, 130, 143, 156, 169, 182, 195 +43	Blanket catch swallows failures and leaks exception messages	Remove catch or log+rethrow; never return a success-shaped payload
app/Http/Controllers/Ivr/BusinessHoursStoreController.php	52, 65, 78, 91, 104, 117, 130, 143, 156, 169, 182, 195 +43	Blanket catch swallows failures and leaks exception messages	Remove catch or log+rethrow; never return a success-shaped payload
app/Http/Controllers/Ivr/BusinessHoursSyncController.php	52, 65, 78, 91, 104, 117, 130, 143, 156, 169, 182, 195 +43	Blanket catch swallows failures and leaks exception messages	Remove catch or log+rethrow; never return a success-shaped payload
app/Http/Controllers/Ivr/BusinessHoursUpdateController.php	52, 65, 78, 91, 104, 117, 130, 143, 156, 169, 182, 195 +43	Blanket catch swallows failures and leaks exception messages	Remove catch or log+rethrow; never return a success-shaped payload
app/Http/Controllers/Ivr/CallAnalyticsDestroyController.php	52, 65, 78, 91, 104, 117, 130, 143, 156, 169, 182, 195 +43	Blanket catch swallows failures and leaks exception messages	Remove catch or log+rethrow; never return a success-shaped payload
app/Http/Controllers/Ivr/CallAnalyticsExportController.php	52, 65, 78, 91, 104, 117, 130, 143, 156, 169, 182, 195 +43	Blanket catch swallows failures and leaks exception messages	Remove catch or log+rethrow; never return a success-shaped payload
app/Http/Controllers/Ivr/CallAnalyticsImportController.php	52, 65, 78, 91, 104, 117, 130, 143, 156, 169, 182, 195 +43	Blanket catch swallows failures and leaks exception messages	Remove catch or log+rethrow; never return a success-shaped payload
app/Http/Controllers/Ivr/CallAnalyticsIndexController.php	52, 65, 78, 91, 104, 117, 130, 143, 156, 169, 182, 195 +43	Blanket catch swallows failures and leaks exception messages	Remove catch or log+rethrow; never return a success-shaped payload
app/Http/Controllers/Ivr/CallAnalyticsStoreController.php	52, 65, 78, 91, 104, 117, 130, 143, 156, 169, 182, 195 +43	Blanket catch swallows failures and leaks exception messages	Remove catch or log+rethrow; never return a success-shaped payload
app/Http/Controllers/Ivr/CallAnalyticsSyncController.php	52, 65, 78, 91, 104, 117, 130, 143, 156, 169, 182, 195 +43	Blanket catch swallows failures and leaks exception messages	Remove catch or log+rethrow; never return a success-shaped payload
app/Http/Controllers/Ivr/CallAnalyticsUpdateController.php	52, 65, 78, 91, 104, 117, 130, 143, 156, 169, 182, 195 +43	Blanket catch swallows failures and leaks exception messages	Remove catch or log+rethrow; never return a success-shaped payload
app/Http/Controllers/Ivr/CallFlowDestroyController.php	52, 65, 78, 91, 104, 117, 130, 143, 156, 169, 182, 195 +43	Blanket catch swallows failures and leaks exception messages	Remove catch or log+rethrow; never return a success-shaped payload
app/Http/Controllers/Ivr/CallFlowExportController.php	52, 65, 78, 91, 104, 117, 130,	Blanket catch swallows failures and leaks exception messages	Remove catch or log+rethrow; never return a success-shaped payload

	143, 156, 169, 182, 195 +43		
<code>app/Http/Controllers/Ivr/CallFlowImportController.php</code>	52, 65, 78, 91, 104, 117, 130, 143, 156, 169, 182, 195 +43	Blanket catch swallows failures and leaks exception messages	Remove catch or log+rethrow; never return a success-shaped payload
<code>app/Http/Controllers/Ivr/CallFlowIndexController.php</code>	52, 65, 78, 91, 104, 117, 130, 143, 156, 169, 182, 195 +43	Blanket catch swallows failures and leaks exception messages	Remove catch or log+rethrow; never return a success-shaped payload
<code>app/Http/Controllers/Ivr/CallFlowStoreController.php</code>	52, 65, 78, 91, 104, 117, 130, 143, 156, 169, 182, 195 +43	Blanket catch swallows failures and leaks exception messages	Remove catch or log+rethrow; never return a success-shaped payload
<code>app/Http/Controllers/Ivr/CallFlowSyncController.php</code>	52, 65, 78, 91, 104, 117, 130, 143, 156, 169, 182, 195 +43	Blanket catch swallows failures and leaks exception messages	Remove catch or log+rethrow; never return a success-shaped payload
<code>app/Http/Controllers/Ivr/CallFlowUpdateController.php</code>	52, 65, 78, 91, 104, 117, 130, 143, 156, 169, 182, 195 +43	Blanket catch swallows failures and leaks exception messages	Remove catch or log+rethrow; never return a success-shaped payload
<code>app/Http/Controllers/Ivr/CallRecordingDestroyController.php</code>	52, 65, 78, 91, 104, 117, 130, 143, 156, 169, 182, 195 +43	Blanket catch swallows failures and leaks exception messages	Remove catch or log+rethrow; never return a success-shaped payload
<code>app/Http/Controllers/Ivr/CallRecordingExportController.php</code>	52, 65, 78, 91, 104, 117, 130, 143, 156, 169, 182, 195 +43	Blanket catch swallows failures and leaks exception messages	Remove catch or log+rethrow; never return a success-shaped payload
<code>app/Http/Controllers/Ivr/CallRecordingImportController.php</code>	52, 65, 78, 91, 104, 117, 130, 143, 156, 169, 182, 195 +43	Blanket catch swallows failures and leaks exception messages	Remove catch or log+rethrow; never return a success-shaped payload
<code>app/Http/Controllers/Ivr/CallRecordingIndexController.php</code>	52, 65, 78, 91, 104, 117, 130, 143, 156, 169, 182, 195 +43	Blanket catch swallows failures and leaks exception messages	Remove catch or log+rethrow; never return a success-shaped payload
<code>app/Http/Controllers/Ivr/CallRecordingStoreController.php</code>	52, 65, 78, 91, 104, 117, 130, 143, 156, 169, 182, 195 +43	Blanket catch swallows failures and leaks exception messages	Remove catch or log+rethrow; never return a success-shaped payload
<code>app/Http/Controllers/Ivr/CallRecordingSyncController.php</code>	52, 65, 78, 91, 104, 117, 130, 143, 156, 169, 182, 195 +43	Blanket catch swallows failures and leaks exception messages	Remove catch or log+rethrow; never return a success-shaped payload
<code>app/Http/Controllers/Ivr/CallRecordingUpdateController.php</code>	52, 65, 78, 91, 104, 117, 130, 143, 156, 169, 182, 195 +43	Blanket catch swallows failures and leaks exception messages	Remove catch or log+rethrow; never return a success-shaped payload
<code>app/Http/Controllers/Ivr/CallRoutingDestroyController.php</code>	52, 65, 78, 91, 104, 117, 130, 143, 156, 169, 182, 195 +43	Blanket catch swallows failures and leaks exception messages	Remove catch or log+rethrow; never return a success-shaped payload
<code>app/Http/Controllers/Ivr/CallRoutingExportController.php</code>	52, 65, 78, 91, 104, 117, 130, 143, 156, 169, 182, 195 +43	Blanket catch swallows failures and leaks exception messages	Remove catch or log+rethrow; never return a success-shaped payload

app/Http/Controllers/Ivr/CallRoutingImportController.php	52, 65, 78, 91, 104, 117, 130, 143, 156, 169, 182, 195 +43	Blanket catch swallows failures and leaks exception messages	Remove catch or log+rethrow; never return a success-shaped payload
app/Http/Controllers/Ivr/CallRoutingIndexController.php	52, 65, 78, 91, 104, 117, 130, 143, 156, 169, 182, 195 +43	Blanket catch swallows failures and leaks exception messages	Remove catch or log+rethrow; never return a success-shaped payload
app/Http/Controllers/Ivr/CallRoutingStoreController.php	52, 65, 78, 91, 104, 117, 130, 143, 156, 169, 182, 195 +43	Blanket catch swallows failures and leaks exception messages	Remove catch or log+rethrow; never return a success-shaped payload
app/Http/Controllers/Ivr/CallRoutingSyncController.php	52, 65, 78, 91, 104, 117, 130, 143, 156, 169, 182, 195 +43	Blanket catch swallows failures and leaks exception messages	Remove catch or log+rethrow; never return a success-shaped payload
app/Http/Controllers/Ivr/CallRoutingUpdateController.php	52, 65, 78, 91, 104, 117, 130, 143, 156, 169, 182, 195 +43	Blanket catch swallows failures and leaks exception messages	Remove catch or log+rethrow; never return a success-shaped payload
app/Http/Controllers/Ivr/Concerns/LoadsIvrModuleData.php	225	Blanket catch swallows failures and leaks exception messages	Remove catch or log+rethrow; never return a success-shaped payload
app/Http/Controllers/Ivr/CustomerProfileIndexController.php	52, 65, 78, 91, 104, 117, 130, 143, 156, 169, 182, 195 +43	Blanket catch swallows failures and leaks exception messages	Remove catch or log+rethrow; never return a success-shaped payload
app/Http/Controllers/Ivr/CustomerProfileStoreController.php	52, 65, 78, 91, 104, 117, 130, 143, 156, 169, 182, 195 +43	Blanket catch swallows failures and leaks exception messages	Remove catch or log+rethrow; never return a success-shaped payload
app/Http/Controllers/Ivr/CustomerProfileUpdateController.php	52, 65, 78, 91, 104, 117, 130, 143, 156, 169, 182, 195 +43	Blanket catch swallows failures and leaks exception messages	Remove catch or log+rethrow; never return a success-shaped payload
app/Http/Controllers/Ivr/DidInventoryDestroyController.php	52, 65, 78, 91, 104, 117, 130, 143, 156, 169, 182, 195 +43	Blanket catch swallows failures and leaks exception messages	Remove catch or log+rethrow; never return a success-shaped payload
app/Http/Controllers/Ivr/DidInventoryExportController.php	52, 65, 78, 91, 104, 117, 130, 143, 156, 169, 182, 195 +43	Blanket catch swallows failures and leaks exception messages	Remove catch or log+rethrow; never return a success-shaped payload
app/Http/Controllers/Ivr/DidInventoryImportController.php	52, 65, 78, 91, 104, 117, 130, 143, 156, 169, 182, 195 +43	Blanket catch swallows failures and leaks exception messages	Remove catch or log+rethrow; never return a success-shaped payload
app/Http/Controllers/Ivr/DidInventoryIndexController.php	52, 65, 78, 91, 104, 117, 130, 143, 156, 169, 182, 195 +43	Blanket catch swallows failures and leaks exception messages	Remove catch or log+rethrow; never return a success-shaped payload
app/Http/Controllers/Ivr/DidInventoryStoreController.php	52, 65, 78, 91, 104, 117, 130, 143, 156, 169, 182, 195 +43	Blanket catch swallows failures and leaks exception messages	Remove catch or log+rethrow; never return a success-shaped payload
app/Http/Controllers/Ivr/DidInventorySyncController.php	52, 65, 78, 91, 104, 117, 130, 143, 156, 169, 182, 195 +43	Blanket catch swallows failures and leaks exception messages	Remove catch or log+rethrow; never return a success-shaped payload

<code>app/Http/Controllers/Ivr/DidInventoryUpdateController.php</code>	52, 65, 78, 91, 104, 117, 130, 143, 156, 169, 182, 195 +43	Blanket catch swallows failures and leaks exception messages	Remove catch or log+rethrow; never return a success-shaped payload
<code>app/Http/Controllers/Ivr/HistoricalReportsDestroyController.php</code>	52, 65, 78, 91, 104, 117, 130, 143, 156, 169, 182, 195 +43	Blanket catch swallows failures and leaks exception messages	Remove catch or log+rethrow; never return a success-shaped payload
<code>app/Http/Controllers/Ivr/HistoricalReportsExportController.php</code>	52, 65, 78, 91, 104, 117, 130, 143, 156, 169, 182, 195 +43	Blanket catch swallows failures and leaks exception messages	Remove catch or log+rethrow; never return a success-shaped payload
<code>app/Http/Controllers/Ivr/HistoricalReportsImportController.php</code>	52, 65, 78, 91, 104, 117, 130, 143, 156, 169, 182, 195 +43	Blanket catch swallows failures and leaks exception messages	Remove catch or log+rethrow; never return a success-shaped payload
<code>app/Http/Controllers/Ivr/HistoricalReportsIndexController.php</code>	52, 65, 78, 91, 104, 117, 130, 143, 156, 169, 182, 195 +43	Blanket catch swallows failures and leaks exception messages	Remove catch or log+rethrow; never return a success-shaped payload
<code>app/Http/Controllers/Ivr/HistoricalReportsStoreController.php</code>	52, 65, 78, 91, 104, 117, 130, 143, 156, 169, 182, 195 +43	Blanket catch swallows failures and leaks exception messages	Remove catch or log+rethrow; never return a success-shaped payload
<code>app/Http/Controllers/Ivr/HistoricalReportsSyncController.php</code>	52, 65, 78, 91, 104, 117, 130, 143, 156, 169, 182, 195 +43	Blanket catch swallows failures and leaks exception messages	Remove catch or log+rethrow; never return a success-shaped payload
<code>app/Http/Controllers/Ivr/HistoricalReportsUpdateController.php</code>	52, 65, 78, 91, 104, 117, 130, 143, 156, 169, 182, 195 +43	Blanket catch swallows failures and leaks exception messages	Remove catch or log+rethrow; never return a success-shaped payload
<code>app/Http/Controllers/Ivr/LiveMonitoringDestroyController.php</code>	52, 65, 78, 91, 104, 117, 130, 143, 156, 169, 182, 195 +43	Blanket catch swallows failures and leaks exception messages	Remove catch or log+rethrow; never return a success-shaped payload
<code>app/Http/Controllers/Ivr/LiveMonitoringExportController.php</code>	52, 65, 78, 91, 104, 117, 130, 143, 156, 169, 182, 195 +43	Blanket catch swallows failures and leaks exception messages	Remove catch or log+rethrow; never return a success-shaped payload
<code>app/Http/Controllers/Ivr/LiveMonitoringImportController.php</code>	52, 65, 78, 91, 104, 117, 130, 143, 156, 169, 182, 195 +43	Blanket catch swallows failures and leaks exception messages	Remove catch or log+rethrow; never return a success-shaped payload
<code>app/Http/Controllers/Ivr/LiveMonitoringIndexController.php</code>	52, 65, 78, 91, 104, 117, 130, 143, 156, 169, 182, 195 +43	Blanket catch swallows failures and leaks exception messages	Remove catch or log+rethrow; never return a success-shaped payload
<code>app/Http/Controllers/Ivr/LiveMonitoringStoreController.php</code>	52, 65, 78, 91, 104, 117, 130, 143, 156, 169, 182, 195 +43	Blanket catch swallows failures and leaks exception messages	Remove catch or log+rethrow; never return a success-shaped payload
<code>app/Http/Controllers/Ivr/LiveMonitoringSyncController.php</code>	52, 65, 78, 91, 104, 117, 130, 143, 156, 169, 182, 195 +43	Blanket catch swallows failures and leaks exception messages	Remove catch or log+rethrow; never return a success-shaped payload
<code>app/Http/Controllers/Ivr/LiveMonitoringUpdateController.php</code>	52, 65, 78, 91, 104, 117, 130,	Blanket catch swallows failures and leaks exception messages	Remove catch or log+rethrow; never return a success-shaped payload

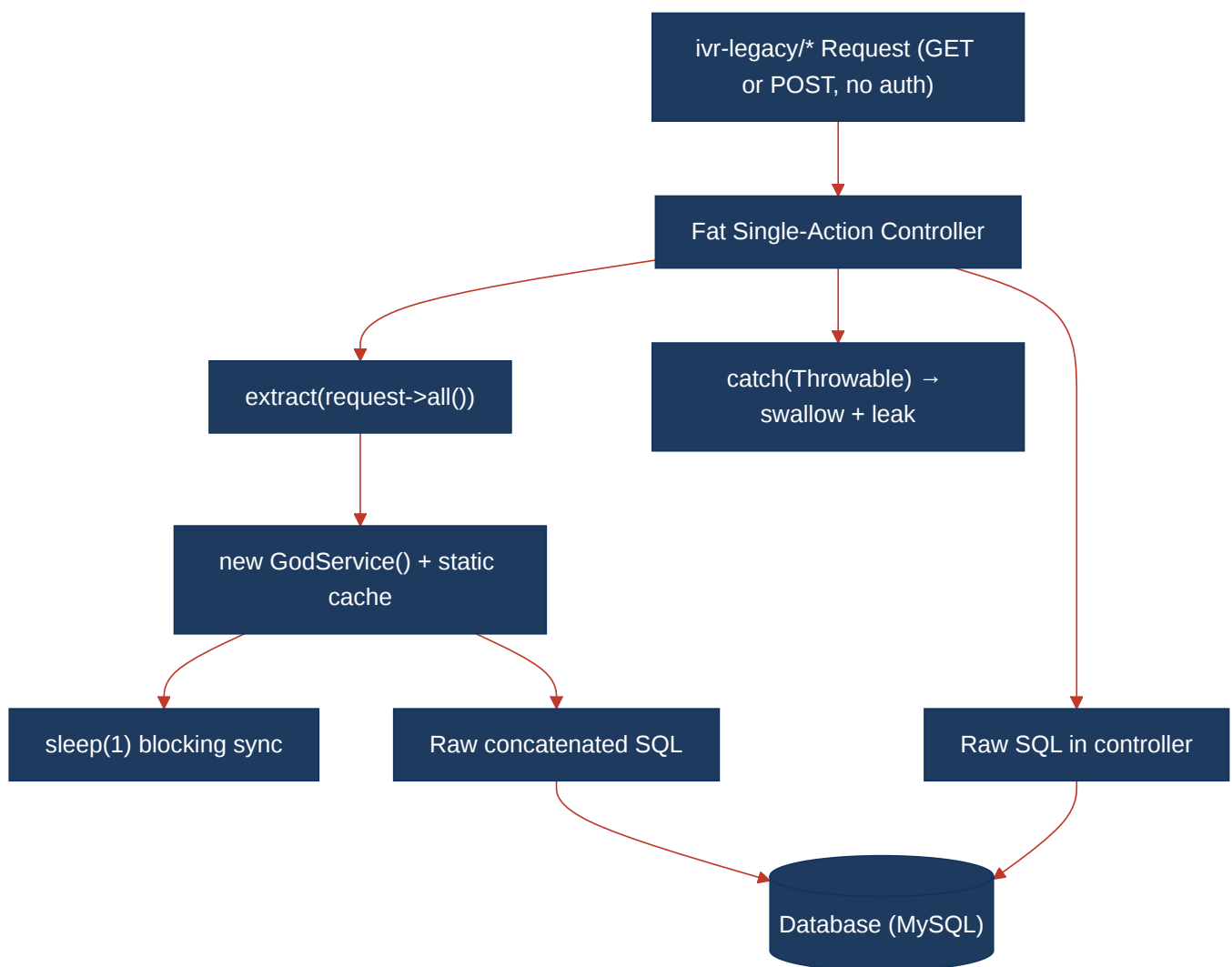
	143, 156, 169, 182, 195 +43		
<code>app/Http/Controllers/Ivr/PromptLibraryDestroyController.php</code>	52, 65, 78, 91, 104, 117, 130, 143, 156, 169, 182, 195 +43	Blanket catch swallows failures and leaks exception messages	Remove catch or log+rethrow; never return a success-shaped payload
<code>app/Http/Controllers/Ivr/PromptLibraryExportController.php</code>	52, 65, 78, 91, 104, 117, 130, 143, 156, 169, 182, 195 +43	Blanket catch swallows failures and leaks exception messages	Remove catch or log+rethrow; never return a success-shaped payload
<code>app/Http/Controllers/Ivr/PromptLibraryImportController.php</code>	52, 65, 78, 91, 104, 117, 130, 143, 156, 169, 182, 195 +43	Blanket catch swallows failures and leaks exception messages	Remove catch or log+rethrow; never return a success-shaped payload
<code>app/Http/Controllers/Ivr/PromptLibraryIndexController.php</code>	52, 65, 78, 91, 104, 117, 130, 143, 156, 169, 182, 195 +43	Blanket catch swallows failures and leaks exception messages	Remove catch or log+rethrow; never return a success-shaped payload
<code>app/Http/Controllers/Ivr/PromptLibraryStoreController.php</code>	52, 65, 78, 91, 104, 117, 130, 143, 156, 169, 182, 195 +43	Blanket catch swallows failures and leaks exception messages	Remove catch or log+rethrow; never return a success-shaped payload
<code>app/Http/Controllers/Ivr/PromptLibrarySyncController.php</code>	52, 65, 78, 91, 104, 117, 130, 143, 156, 169, 182, 195 +43	Blanket catch swallows failures and leaks exception messages	Remove catch or log+rethrow; never return a success-shaped payload
<code>app/Http/Controllers/Ivr/PromptLibraryUpdateController.php</code>	52, 65, 78, 91, 104, 117, 130, 143, 156, 169, 182, 195 +43	Blanket catch swallows failures and leaks exception messages	Remove catch or log+rethrow; never return a success-shaped payload
<code>app/Http/Controllers/Ivr/QueueManagementDestroyController.php</code>	52, 65, 78, 91, 104, 117, 130, 143, 156, 169, 182, 195 +43	Blanket catch swallows failures and leaks exception messages	Remove catch or log+rethrow; never return a success-shaped payload
<code>app/Http/Controllers/Ivr/QueueManagementExportController.php</code>	52, 65, 78, 91, 104, 117, 130, 143, 156, 169, 182, 195 +43	Blanket catch swallows failures and leaks exception messages	Remove catch or log+rethrow; never return a success-shaped payload
<code>app/Http/Controllers/Ivr/QueueManagementImportController.php</code>	52, 65, 78, 91, 104, 117, 130, 143, 156, 169, 182, 195 +43	Blanket catch swallows failures and leaks exception messages	Remove catch or log+rethrow; never return a success-shaped payload
<code>app/Http/Controllers/Ivr/QueueManagementIndexController.php</code>	52, 65, 78, 91, 104, 117, 130, 143, 156, 169, 182, 195 +43	Blanket catch swallows failures and leaks exception messages	Remove catch or log+rethrow; never return a success-shaped payload
<code>app/Http/Controllers/Ivr/QueueManagementStoreController.php</code>	52, 65, 78, 91, 104, 117, 130, 143, 156, 169, 182, 195 +43	Blanket catch swallows failures and leaks exception messages	Remove catch or log+rethrow; never return a success-shaped payload
<code>app/Http/Controllers/Ivr/QueueManagementSyncController.php</code>	52, 65, 78, 91, 104, 117, 130, 143, 156, 169, 182, 195 +43	Blanket catch swallows failures and leaks exception messages	Remove catch or log+rethrow; never return a success-shaped payload
<code>app/Http/Controllers/Ivr/QueueManagementUpdateController.php</code>	52, 65, 78, 91, 104, 117, 130, 143, 156, 169, 182, 195 +43	Blanket catch swallows failures and leaks exception messages	Remove catch or log+rethrow; never return a success-shaped payload

Not observed (rated Good): H10 (schema — FK columns indexed via `->index()` / `->constrained()` , every migration has a reversible `down()` , no drift in tracked migrations); H15 (dependencies — Laravel 11 / Sanctum 4 are current, `roave/security-advisories` dev-dependency blocks installing known-vulnerable packages; a live `composer audit` could not run in the offline snapshot, so 0 CVEs is "not measured, none observed").

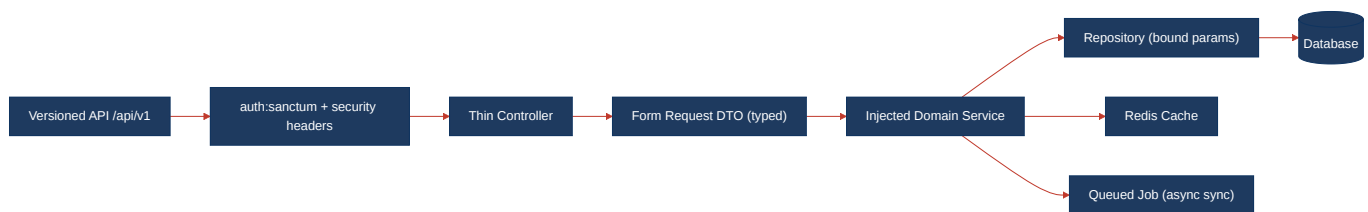
Not applicable — no API surface issues suppressed: an API surface *does* exist (`routes/api.php` + `ivr-legacy`), so H6–H7 are assessed above rather than marked N/A.

4.3 Diagrams

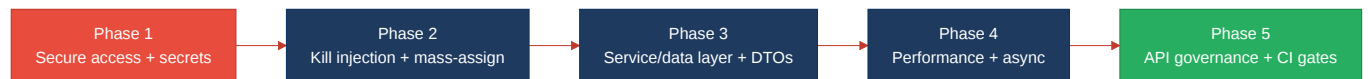
Current backend request path



Modernized service-layer target



Improvement roadmap



4.4 Actions Required

Hotspot	Action	Rating	Priority
H12 Auth & Authorization	Apply <code>auth:sanctum</code> to all 80 <code>ivr-legacy</code> routes; replace hard-coded <code>tenant_id=1</code> with authenticated tenant + object-level policies	HIGH RISK	CRITICAL
H13 Security Vulnerabilities	Bind all SQL parameters; replace <code>\$guarded=[]</code> with <code>\$fillable</code> ; ship <code>APP_DEBUG=false</code> ; stop returning exception messages	HIGH RISK	CRITICAL
H16 Secrets in Source	Move <code>config/ivr_legacy.php</code> + 12 service keys to secrets manager and rotate; add secret-scanning to CI	HIGH RISK	CRITICAL
H1 Dynamic Variable Creation	Replace <code>extract()</code> in 92 files with typed Form Request DTOs and explicit field access	HIGH RISK	CRITICAL
H3 Direct SQL Outside Data Layer	Route all <code>DB::</code> access through repositories with bound params; add arch rule banning <code>DB::</code> in controllers	HIGH RISK	HIGH
H5 Missing Service Layer	Extract cohesive domain services; thin the 82 controllers to DTO → service → response	HIGH RISK	HIGH
H4 Static / Singleton Abuse	Bind services in container; inject instead of <code>new</code> ; collapse static helpers	HIGH RISK	HIGH
H2 Global Mutable State	Remove <code>public static \$sharedRuntimeCache</code> ; use scoped cache	HIGH RISK	HIGH
H14 Performance & Caching	Delete <code>sleep(1)</code> (queue it); eliminate N+1 accessors; add Redis + <code>Cache-Control</code>	HIGH RISK	HIGH
H18 Swallowed Exceptions	Remove blanket <code>catch(\Throwable)</code> that swallows/leaks; log+rethrow with correlation IDs	HIGH RISK	HIGH
H6 API Sprawl	Collapse GET+POST duplicate routes into versioned RESTful resources	HIGH RISK	HIGH
H7 API Governance	Add OpenAPI spec, Spectral linting, contract tests and <code>/v1</code> versioning in CI	HIGH RISK	HIGH
H8 Weak Architecture	Enforce HTTP → domain → data layering; move model queries to repositories; add arch tests	HIGH RISK	HIGH

H11 Middleware Weakness	Add security-headers + correlation-id logging; explicit CORS; remove IP auth-bypass	MODERATE	MEDIUM
H9 Module Inventory	Delete/archive 17 unreferenced dead files; document module public APIs	MODERATE	MEDIUM
H17 Code Quality	Raise PHPStan past level 1; add duplication/complexity gates; collapse cloned methods	MODERATE	MEDIUM

4.5 Expected Outcomes

- **Access control restored:** guarding the 80 `ivr-legacy` routes and deriving tenant from the authenticated user eliminates the unauthenticated-mutation and IDOR exposure that currently lets any caller read/write any tenant's data.
- **Injection eliminated:** parameter binding across controllers and repositories, plus `$fillable` allow-lists, removes the SQL-injection and mass-assignment surface (OWASP A03/A08).
- **Credentials secured:** moving 15+ hardcoded secrets to a vault and rotating them prevents lateral movement into Salesforce and blocks future leakage via CI secret scanning.
- **Typed, traceable data flow:** replacing `extract()` with Form Request DTOs makes request handling type-checkable and lets PHPStan rise above level 1.
- **Reusable domain layer:** a real service + repository layer lets IVR logic be shared across HTTP, CLI and queued jobs, and shrinks the 82 fat controllers to thin delegators.
- **Performance headroom:** removing 540 blocking `sleep(1)` calls, eliminating 420 N+1 accessors and adding Redis caching frees the worker pool and cuts database load under concurrency.
- **Observable failures:** removing blanket `catch(\Throwable)` swallowing and adding correlation-ID logging surfaces production errors to monitoring instead of hiding them behind HTTP 200s.
- **Contract stability:** an OpenAPI spec, versioned routes, API linting and contract tests in CI stop breaking changes from reaching integrators undetected.